

Operating System Fundamentals

Part 1



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How to complete this module

This module covers practical's using Cisco Packet Tracer, GoToMyPC/LOD and access to Miro

Please ensure you have these installed and working on your own machine.

If you have any issues, please inform your tutor.

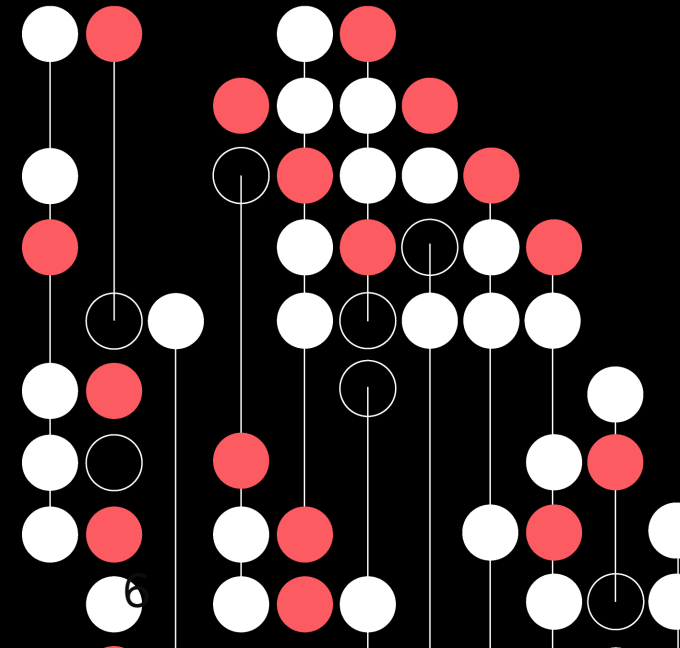


How to complete this module

As part of your live event you will have the slides and activity book to complete.

- Open your '**Learner Activity Document**' for this course and use it to document your answers.
- Add your **full name and date** in the '**header**' to your **Learner Activity Document**.
- Use this PowerPoint as a guide to take you through the course.
- You may need to ask/do some research to answer some of the criteria.
- Once your **Learner Activity Document** is completed, upload it to **BUD** along with your **Certificate of Completion**.
- And most importantly, if you need **help message or email your tutor**.

Computer components



Computer components

Computers play a crucial role in contemporary life, facilitating everything from personal communication to extensive data processing.

Central to every computer is a set of vital components that collaborate to execute its functions.

Gaining knowledge about these components sheds light on the workings of computers and underlines the significance of each element within the system.

Computer components can generally be categorised into two main types: **hardware and software**.

Hardware encompasses the tangible parts of the computer, while software consists of the instructions and data that the hardware processes.

Computer components: Key hardware components

Key hardware components that you will investigate:

- Central Processing Unit (CPU)
- Memory (RAM)
- Storage (HDD/SSD)
- Motherboard
- Power Supply Unit (PSU)
- Input/Output devices
- Graphics Processing Unit (GPU)
- Cooling systems
- Cases
- Build types

Computer components: CPU

The Central Processing Unit (CPU), often referred to as the brain of the computer, is responsible for performing all calculations and data manipulation required by programs.

It contains transistors that function as logic gates, enabling it to process instructions efficiently.

The speed of a processor is measured in gigahertz (GHz), with a 1 GHz processor capable of executing one billion cycles per second.

This incredible speed generates significant heat, requiring effective cooling solutions.

Computer components: CPU manufacturers

Intel and AMD are the two leading manufacturers of Central Processing Units (CPUs).

Both companies offer a wide range of processors designed for different use cases, from basic computing to high-performance gaming and professional workloads.

Computer components: Choosing a CPU

CPU

There are various factors in choosing the right CPU brand and line for your specific computing needs and use case. Some factors include:

Performance requirements: Determine whether your tasks are CPU-intensive (e.g., gaming, video editing, 3D rendering).

Budget constraints: Choose a processor that provides the best value for money within your budget range. AMD often offers higher core counts at lower prices, while Intel frequently excels in premium segments.

Intended use: For gaming, look for high clock speeds and strong single-threaded performance. For professional workloads, prioritise multi-core and multi-threaded processors.

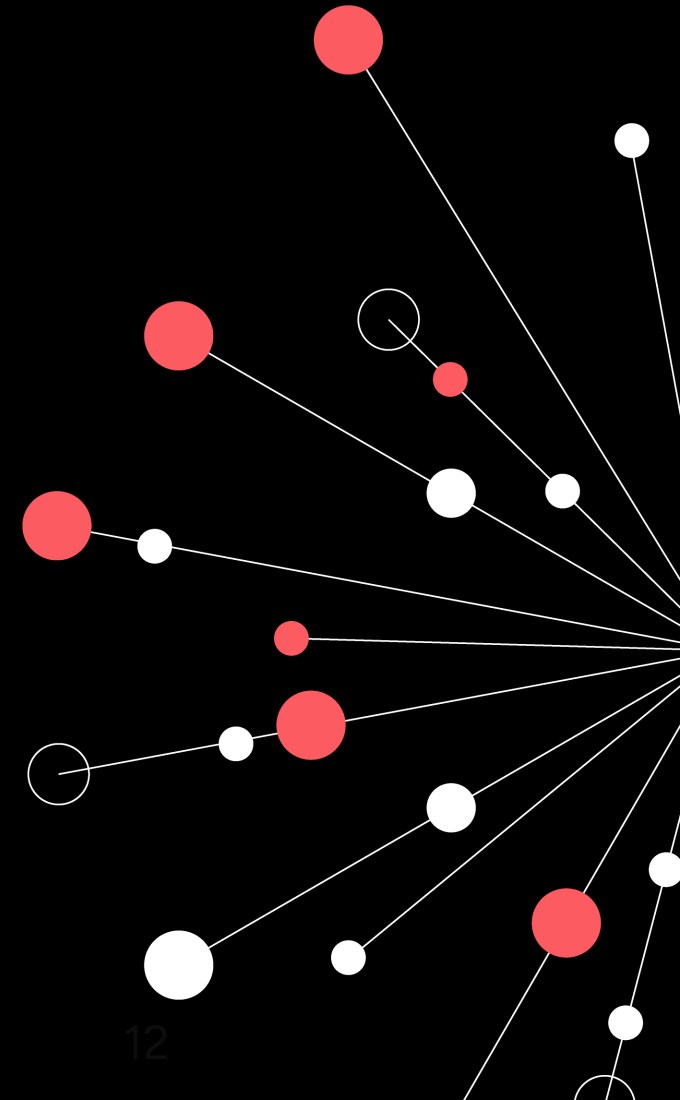
Platform compatibility: Ensure the CPU is compatible with your existing or planned motherboard (socket type and chipset). Consider future upgrade paths and platform longevity.

Power efficiency and heat output: If you prioritise a quieter or energy-efficient system, look for CPUs with lower TDP (Thermal Design Power).

Activity 1: CPU Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: CPUs

Other than PCs, processors can also be found in various other hardware, including smartphones, tablets, gaming consoles, and embedded systems.

These processors are manufactured by companies such as Qualcomm, MediaTek, Nvidia, Apple, and Samsung, which specialise in mobile and custom chip designs.

Computer components: Random Access Memory (RAM)

Random Access Memory (RAM) is a critical component of modern computing, acting as the temporary storage space that allows a computer to perform tasks efficiently.

Unlike long-term storage devices such as hard drives or SSDs, RAM is a type of volatile memory, meaning it loses its data when power is turned off.

Its primary function is to provide fast, short-term access to data and instructions that the CPU and other components need to execute tasks.

Computer components: Types of RAM

Static RAM (SRAM): Stores data using flip-flop circuits, making it faster and more reliable than other types of RAM. Used in cache memory and buffers due to its high speed. More expensive and requires more power than other RAM types.

Dynamic RAM (DRAM): Stores data using capacitors that need constant refreshing to retain information. Slower but more cost-effective and widely used as system memory.

Synchronous DRAM (SDRAM): Synchronises with the CPU clock, enabling faster data transfer. Improved efficiency over earlier asynchronous DRAM.

Computer components: Types of RAM

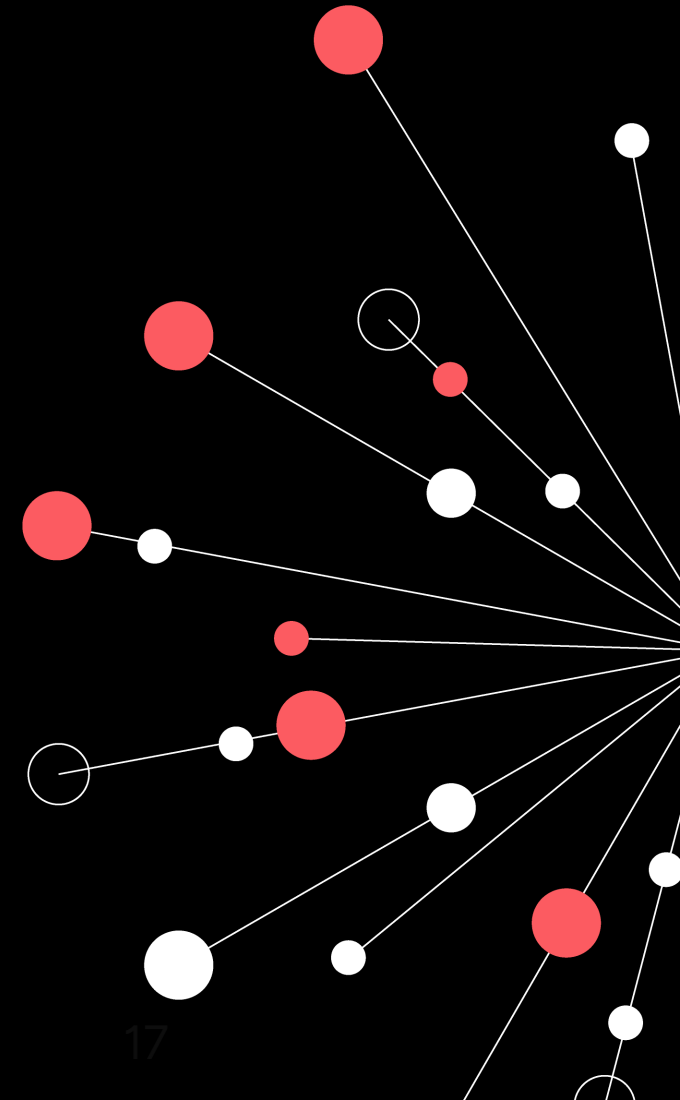
DDR (Double Data Rate) SDRAM: Transfers data on both the rising and falling edges of the clock signal, doubling data throughput. Several types:

- **DDR2:** Doubled the speed of DDR, lower power consumption and higher bandwidth.
- **DDR3:** Increased speed and efficiency, and lower voltage requirements for energy savings.
- **DDR4:** Improved bandwidth and power efficiency and increased capacity per module, suitable for modern computing demands.
- **DDR5:** Higher speeds and bandwidth, lower power consumption per bit transferred, optimised for high-performance tasks like gaming, AI, and data centres.

Activity 2: RAM Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: RAM

The speed of RAM refers to how quickly it can read from or write data to its memory cells, directly influencing a system's overall performance.

RAM speed is typically measured in **megahertz (MHz)**, indicating the number of cycles it can perform per second.

Modern RAM speeds vary by generation:

- **DDR3:** Ranges from 800 MHz to 2133 MHz.
- **DDR4:** Typically ranges from 2133 MHz to 3200 MHz, with overclocked models reaching higher speeds.
- **DDR5:** Starts at 4800 MHz and can go beyond 8000 MHz in high-performance systems.

Computer components: Storage

Modern computers rely on two primary types of storage devices: Solid-State Drives (SSD) and Hard Disk Drives (HDD).

These storage solutions play a critical role in determining a computer's performance, capacity, and overall user experience.

While both serve the purpose of storing data, their underlying technologies, speed, durability, and price points differ significantly.

Each type comes with distinct characteristics, advantages, and limitations that make them better suited for specific applications and user needs.

Computer components: HDDs

HDDs use magnetic platters that spin at high speeds to read and write data using a mechanical arm.

Key features:

- **Capacity:** Offers high storage capacity at a lower cost per gigabyte. Commonly available in sizes ranging from 500 GB to several terabytes (TB).
- **Speed:** Slower than SSDs due to the mechanical nature of data access. Speeds typically range between 80-160 MB/s.
- **Durability:** More prone to physical damage due to moving parts.
- **Price:** Cost-effective, making them ideal for large-scale data storage.

Computer components: SSDs

SSDs use flash memory (non-volatile NAND chips) to store data electronically, with no moving parts.

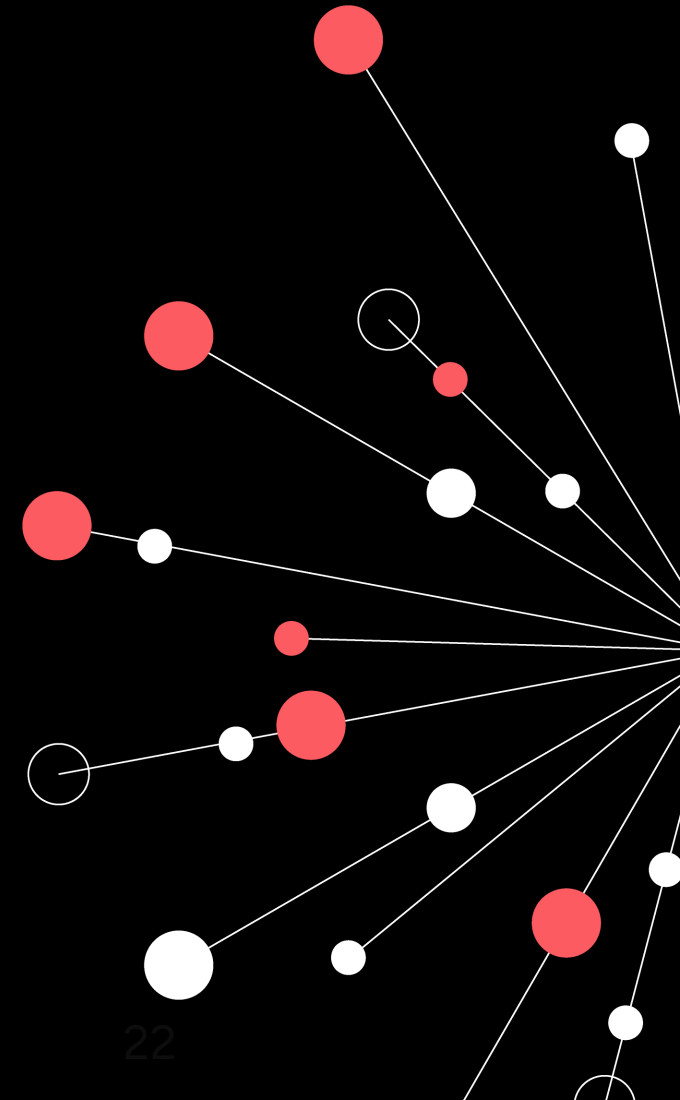
Key features:

- **Speed:** Much faster than HDDs, with read/write speeds ranging from 500 MB/s to several gigabytes per second (especially with NVMe SSDs).
- **Durability:** More resistant to physical damage due to the absence of mechanical components.
- **Form factor:** Compact design, available in 2.5-inch drives, M.2, and PCIe cards.
- **Price:** More expensive per gigabyte compared to HDDs, but prices have been steadily decreasing.

Activity 3: Storage Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

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In summary: Storage

In today's computing world, many systems adopt a hybrid storage approach to leverage the strengths of both types of drives.

SSDs are typically used for operating systems, frequently accessed applications, and high-speed operations, significantly enhancing the overall responsiveness of the system.

Meanwhile, HDDs are employed as secondary storage solutions, ideal for storing large files such as backups, media collections, and less frequently used data.

This combination balances cost and performance, providing users with the benefits of rapid access to important files and the capacity to store vast amounts of data affordably.

Computer components: Motherboards

The motherboard is the central hub of a computer, serving as the main circuit board that connects and coordinates all the hardware components.

Often referred to as the 'backbone' of the system, it provides the pathways for communication between the CPU, RAM, storage devices, graphics cards, and other peripherals.

Computer components: Motherboards

Motherboards come in various sizes and configurations, designed to support different types of processors, memory modules, and expansion cards.

They also house essential components like the chipset, BIOS/UEFI firmware, and input/output ports, which collectively determine the system's compatibility, performance, and upgrade potential.

Computer components: Motherboard sizes

Motherboards are available in various sizes, also known as form factors, which dictate their physical dimensions, layout, and compatibility with computer cases and components.

Choosing the right motherboard size depends on your computing needs, the case size, and the desired expandability of your system.

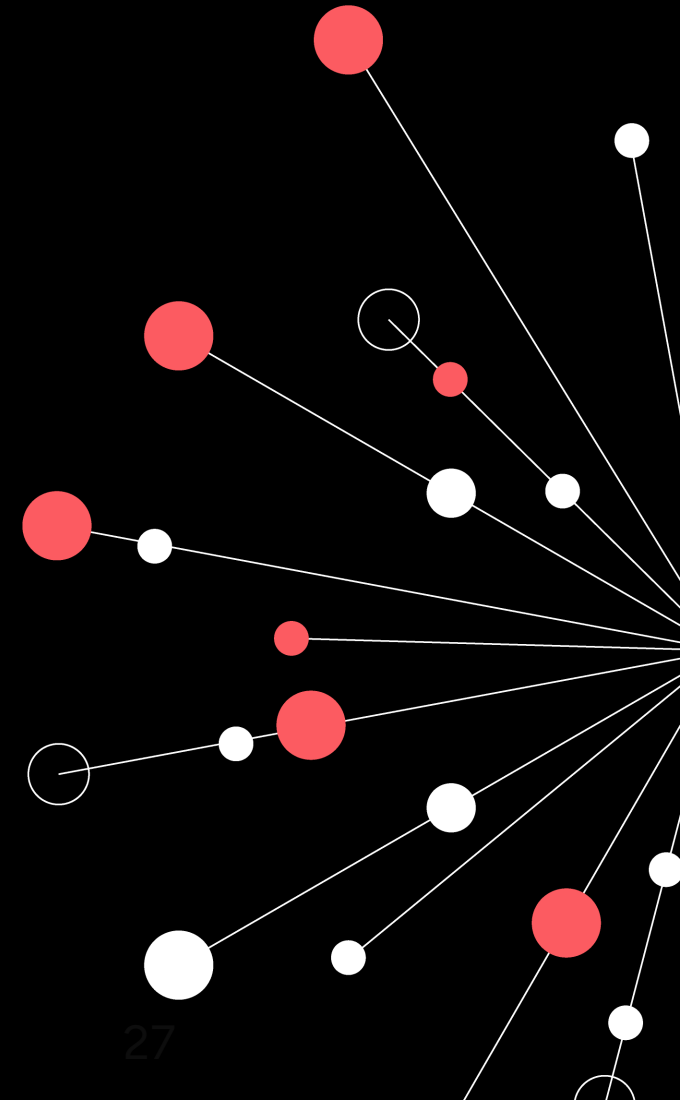
Below are the most common motherboard sizes:

- ATX (Advanced Technology Extended)
- Micro-ATX (mATX)
- Mini-ITX
- E-ATX (Extended ATX)
- Nano-ITX and Pico-ITX

Activity 4: Motherboards Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: Motherboards

Motherboard size impacts the system's expandability, power requirements, and overall design.

While larger boards like ATX and E-ATX offer maximum functionality, smaller sizes like Micro-ATX and Mini-ITX are ideal for compact and budget-friendly builds.

Selecting the right form factor ensures your system meets your specific needs while fitting your chosen case and components.

Computer components: Power Supply Unit (PSU)

The Power Supply Unit (PSU) is a crucial component in any computer system.

It's responsible for converting electrical power from a wall outlet into the appropriate voltage and current needed to power the various internal components of the computer, such as the motherboard, CPU, storage devices, and graphics card.

Without a functioning PSU, a computer cannot operate, as it is the source of power for all other hardware.

Computer components: PSU wattage

The power wattage of a Power Supply Unit (PSU) refers to the maximum amount of electrical power it can supply to a computer's components.

It is typically measured in watts (W) and indicates the total power output of the PSU.

This wattage determines whether the PSU can handle the power requirements of all the hardware components inside the system, such as the CPU, GPU, motherboard, storage devices, and cooling fans.

Computer components: Low wattage and mid-range PSUs

Low-wattage PSUs (under 400W)

Use case: Suitable for basic systems with low power consumption, such as office computers, budget builds, or older desktops.

Mid-range PSUs (400W to 600W)

Use case: Ideal for mainstream systems that include dedicated graphics cards, moderate gaming, and general computing tasks.

Computer components: High and ultra-high wattage PSUs

High-wattage PSUs (600W to 1000W)

Use case: Designed for high-performance systems such as gaming PCs, workstations, or video editing rigs that use powerful CPUs, multiple GPUs, and several storage devices.

Ultra-high-wattage PSUs (1000W and above)

Use case: Required for extreme systems or multi-GPU setups, typically used in servers, cryptocurrency mining rigs, or custom-built systems with multiple high-end components.

Computer components: Choosing the right PSU wattage

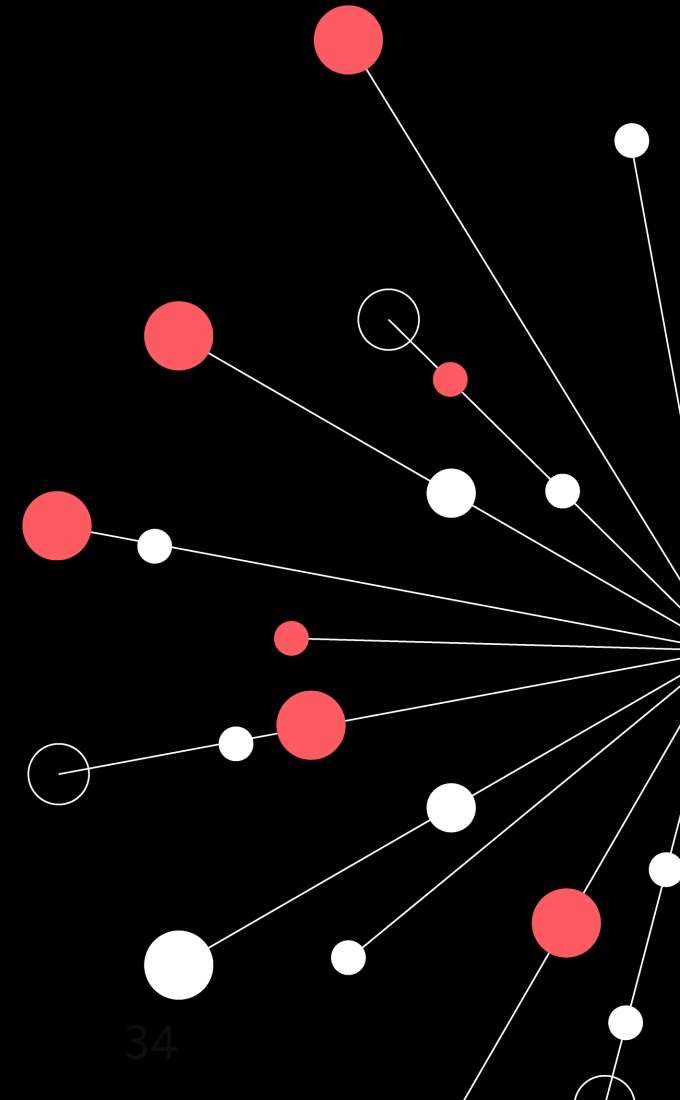
How to choose the right PSU wattage:

- Calculate system power requirements: consider the combined wattage of all components, including the CPU, GPU, storage devices, and cooling fans.
- Leave some headroom: it's a good practice to choose a PSU with 20-30% more wattage than your system's maximum requirement to account for power spikes and future upgrades.
- Efficiency ratings: choose a PSU with a higher 80 plus rating (e.g., 80 plus gold, platinum) for better energy efficiency and reduced heat output.

Activity 5: PSU Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: PSUs

The efficiency rating of a PSU indicates how effectively the PSU converts incoming AC power from the wall outlet into usable DC power for the system's components.

Higher efficiency means less energy is wasted as heat, and the PSU performs better overall.

80 Plus Certification is a widely recognised standard that categorises PSUs based on their efficiency at different loads (typically 20%, 50%, and 100% of their rated capacity).

The certification helps users choose energy-efficient power supplies, which can reduce electricity consumption and heat generation.

Computer components: Input/Output (I/O) devices

Input and Output (I/O) devices are essential for enabling communication between a computer and its user or the external environment.

These devices allow for the exchange of data and commands, facilitating interaction with the system, as well as the retrieval and presentation of processed information.

I/O devices are categorised into input devices, which allow users to send data to the computer, and output devices, which enable the system to present information to the user.

Computer components: Input devices

Input devices are hardware components used to send data to the computer.

They allow users to interact with the system by providing instructions or information that the computer processes. Common examples of input devices include:

- Keyboard
- Mouse
- Touchscreen
- Microphone
- Scanner
- Webcam

Computer components: Output devices

Output devices are hardware components that display or present the data processed by the computer to the user in a human-readable format.

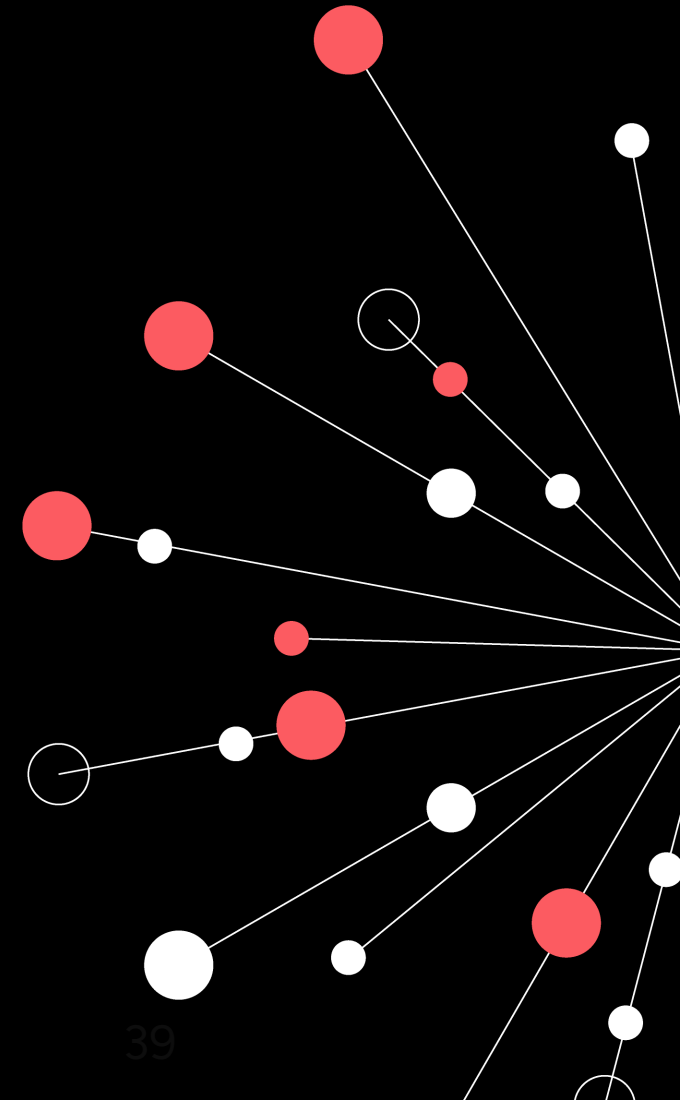
Common examples of output devices include:

- Monitor
- Printer
- Speakers
- Headphones

Activity 6: I/O devices Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: I/O devices

Input and output devices are integral to the functioning of modern computers, as they enable data exchange between the system and the user or external devices.

Input devices allow users to interact with the computer, while output devices present the results of the computer's processing.

Computer components: Graphics Processing Unit (GPU)

A Graphics Processing Unit (GPU) is a specialised electronic circuit designed to accelerate the processing of images, video, and graphics.

While the CPU (Central Processing Unit) handles general-purpose tasks and complex computations, the GPU focuses on parallel processing.

It is particularly suited for rendering visuals, handling graphics-intensive tasks, and performing computations that can be parallelised.

Computer components: GPU characteristics

Unlike CPUs, which are optimised for single-threaded performance, GPUs have thousands of smaller cores designed for parallel processing.

This allows GPUs to execute many tasks simultaneously, making them ideal for tasks such as rendering high-definition graphics, running simulations, and processing large volumes of data.

Modern GPUs are designed with powerful compute capabilities, often used in data centres and for scientific research.

Technologies like CUDA (for NVIDIA GPUs) and OpenCL enable developers to use GPUs for general-purpose computation, speeding up complex algorithms in fields like artificial intelligence, deep learning, and big data analysis.

Computer components: Integrated GPUs

These are built into the same chip as the CPU and share system memory. Integrated GPUs are commonly found in budget laptops and desktops and are suitable for light gaming and everyday tasks.

Computer components: Dedicated GPUs

A dedicated GPU is a separate, stand-alone card with its own memory (VRAM) and processing power.

These GPUs are used in high-performance systems, such as gaming PCs, workstations, and servers.

Major manufacturers of dedicated GPUs include NVIDIA, AMD, and Intel.

Popular lines include NVIDIA's GeForce and RTX series, AMD's Radeon cards, and Intel's Arc Cards.

Computer components: Workstation GPUs

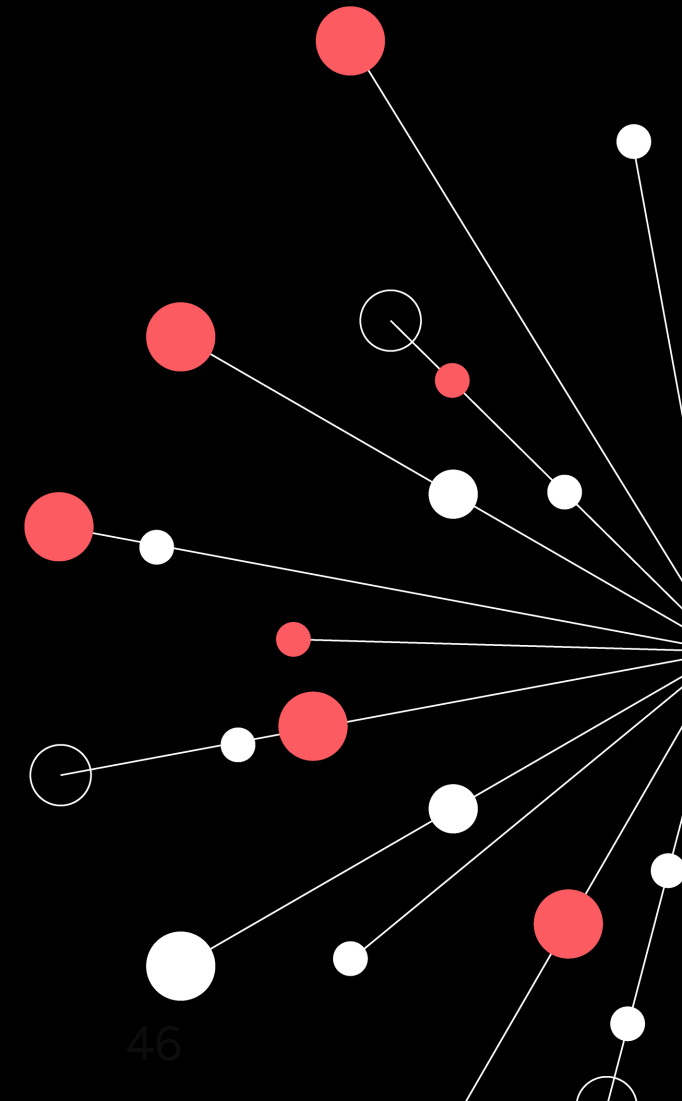
These GPUs are optimised for professional use, such as 3D rendering, CAD (Computer-Aided Design), and scientific computing.

They are typically used in specialised workstations for tasks that require high precision and compute-heavy workloads.

Activity 7: GPUs Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: Graphics Processing Unit (GPU)

The GPU has evolved from a simple graphics rendering device into a powerful and versatile processor that drives much of today's computing, especially in fields requiring massive parallel processing power.

Whether you're gaming, creating content, or working on complex simulations, the GPU plays a vital role in delivering fast, efficient, and high-quality performance.

Choosing the right GPU depends on your needs, from budget-friendly options for basic tasks to high-end models for gaming or professional workloads.

Computer components: Cooling systems

PC cooling methods are essential to ensure that the components of a computer, especially the CPU, GPU, and power supply, operate within safe temperature ranges to prevent overheating and maintain performance.

Overheating can lead to thermal throttling, system instability, or even hardware failure.

Computer components: Types of cooling system

Air cooling:

- The most common and affordable method used in most PCs. It relies on heat sinks and fans to dissipate heat away from the components.

Liquid cooling (also known as water cooling)

- More efficient method for cooling high-performance systems like gaming rigs or workstations. Uses a closed-loop system that circulates coolant (typically a mixture of water and antifreeze)

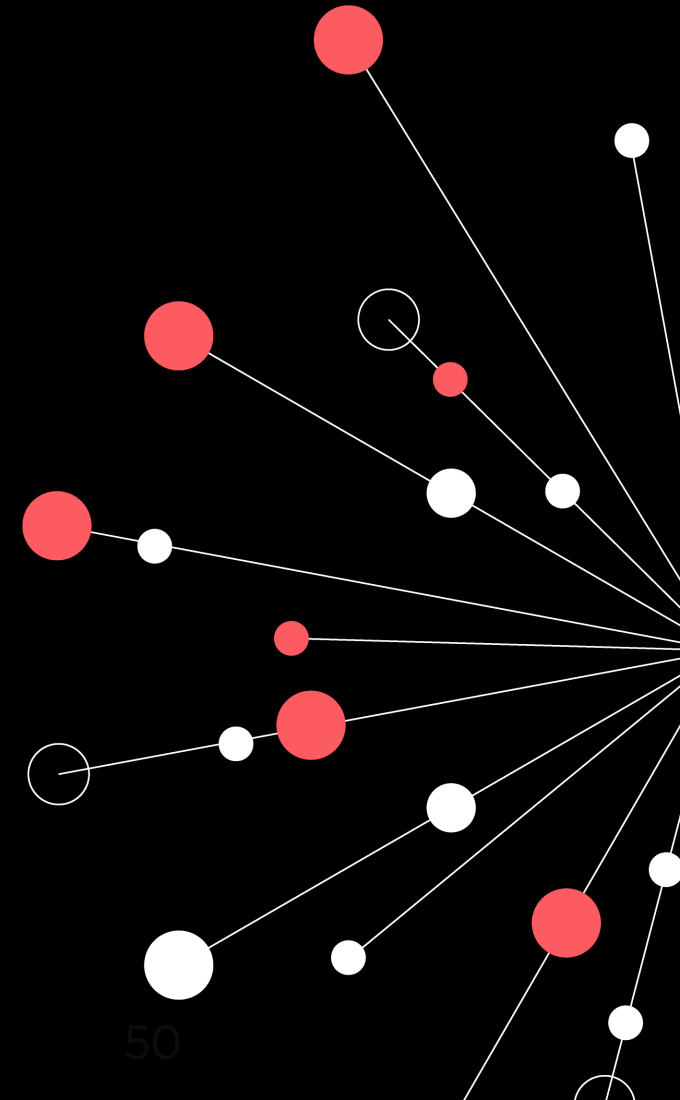
Hybrid cooling

- Combination of air and liquid cooling, often used in specific components, like using a liquid cooler for the CPU but keeping air cooling for the GPU or other parts.

Activity 8: Cooling systems Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: Cooling systems

Selecting the right cooling method depends on the type of PC and its use case.

For everyday use, air cooling and passive cooling are often sufficient, while high-performance gaming PCs or workstations benefit from liquid or even more exotic cooling solutions.

It's essential to balance performance, noise levels, and budget when choosing a cooling method.

Computer components: Cases

A PC case is an enclosure that houses the various components of a computer system.

It serves not only as a protective shell but also as a crucial part of the overall design, functionality, and performance of a PC.

A well-chosen PC case can significantly affect airflow, cooling, aesthetics, and even the ease of building and upgrading your system.

Computer components: Types of PC cases

Full tower: These are the largest PC cases, providing maximum space for additional components, multiple GPU setups, water cooling systems, and numerous storage devices.

Mid tower: These are the most common size for most builds, offering a good balance between space, cooling, and portability.

Mini tower: These are smaller cases designed for compact builds. They are suitable for users with limited space or those building a budget-friendly PC.

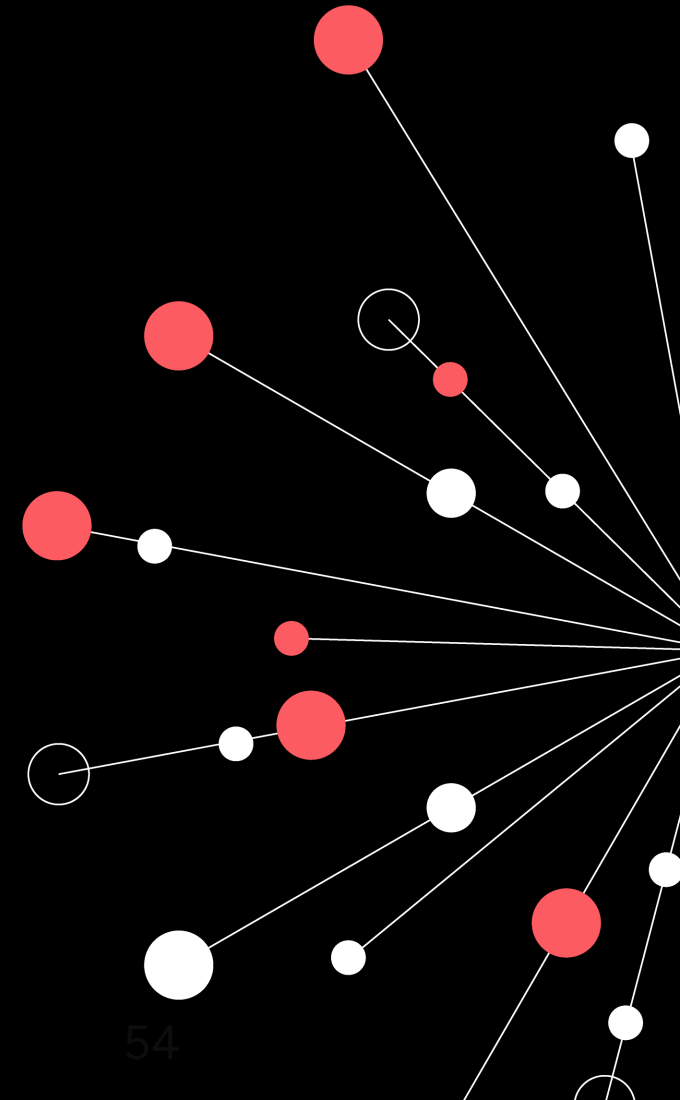
Mini-ITX / SFF (Small Form Factor): These ultra-compact cases are designed for small, portable builds, often used in HTPCs (Home Theatre PCs) or gaming systems.

Cube cases: These are unique in shape, offering a balance between traditional tower and small form factor builds.

Activity 9: PC cases Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: Choosing the right PC case

- **Type of system:** For a gaming rig, airflow and GPU space will be crucial. For a workstation, cable management and space for storage will be more important. For a compact system, focus on size and portability.
- **Component Size:** Make sure your components, such as the motherboard, CPU cooler, GPU, and power supply, fit within the case's dimensions.
- **Budget:** Prices for PC cases vary greatly. Budget-friendly cases can provide decent airflow and features.
- **Expansion and Upgradability:** Consider how much room the case has for future upgrades. If you plan to add more storage drives, GPUs, or cooling solutions later, a case with ample expansion slots and drive bays is crucial.

Computer components: PC builds

The terms **vendor-built PC**, **Pre-built PC**, and **DIY build** are often used in discussions about custom computers, but they refer to different approaches.

Here's a breakdown of each term and the key differences:

- **Vendor-built PC:** Assembled by manufacturer. Typically, higher due to brand markup.
- **Pre-built PC:** Assembled by third-party vendor. Often more expensive than DIY.
- **Building a PC (DIY):** Assembled by the user. Can be cheaper (no markup).

Computer components: Vendor-built PCs

A **vendor-built PC** typically refers to a system assembled and sold by a specific manufacturer or vendor.

These PCs are often made by well-known companies like Dell, HP, Lenovo, Acer, and others.

The vendor handles the selection of components, assembly, and testing, and then sells the finished product as a complete system.

The PC is fully assembled by the manufacturer, and it often comes with pre-installed software and drivers.

Computer components: pre-built PCs

A **pre-built PC** refers to a computer that is fully assembled and sold by a third-party vendor or company, often including a specific focus like gaming or professional workstations.

Unlike vendor-built PCs that are from specific manufacturers, pre-built PCs can come from specialised companies like CyberPowerPC, iBUYPOWER, or Origin PC, which focus on custom-built computers for specific use cases (gaming, video editing, etc.).

Like vendor-built PCs, pre-built systems are assembled and ready to use right out of the box.

Computer components: DIY/Building a PC

DIY/Building a PC refers to the process of selecting, purchasing, and assembling all of the components yourself.

This option allows you to choose every aspect of the system, from the CPU and GPU to the case and cooling solutions.

It's commonly referred to as a **DIY (Do-It-Yourself) build**.

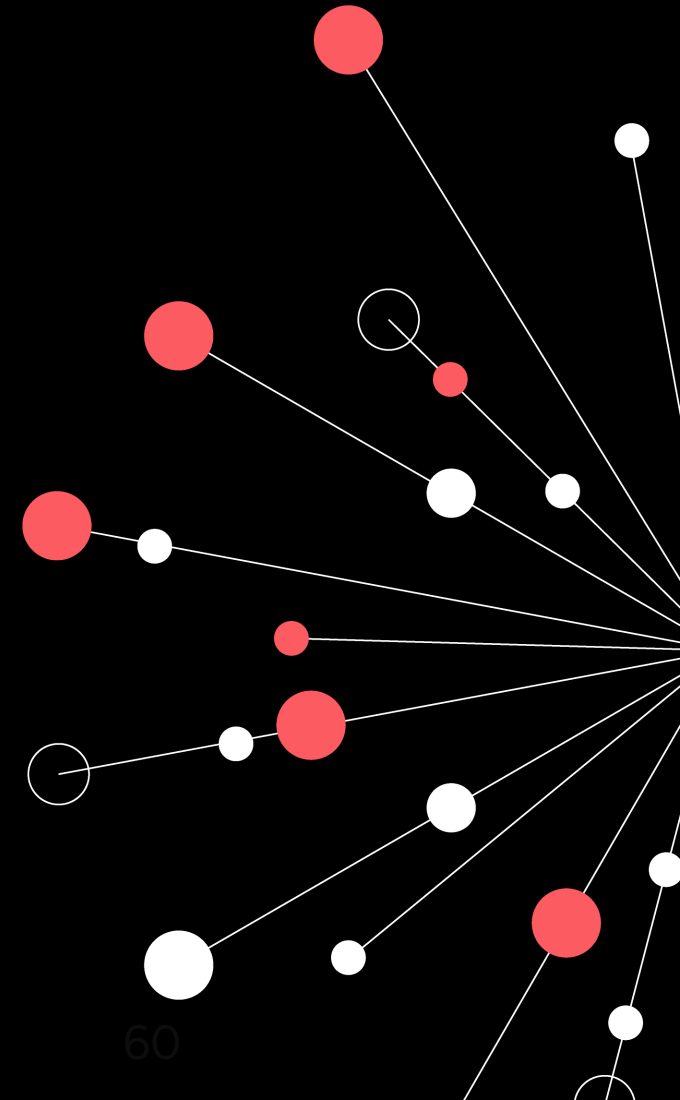
You have complete control over each part of the system, allowing you to select components that best meet your needs.

Building your own PC can be a learning experience, helping you understand the inner workings of your system.

Activity 10: PC builds Q&A

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In summary: PC Builds

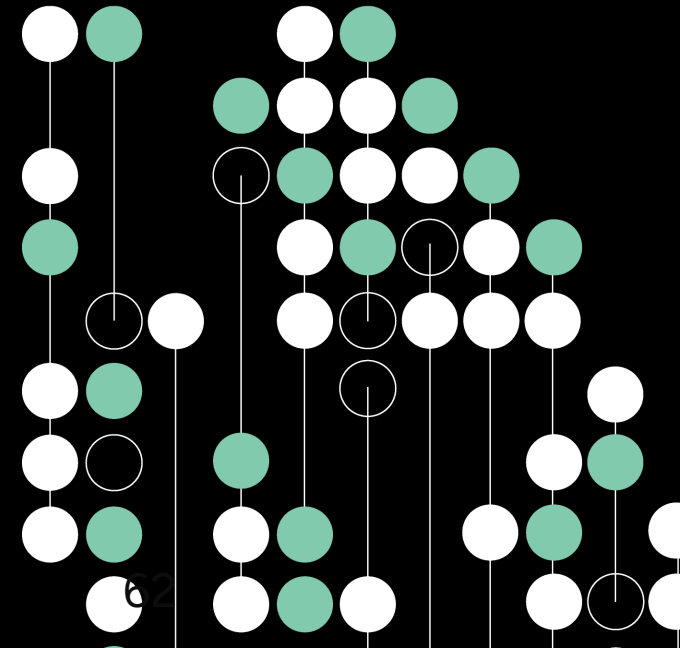
Vendor-built PCs are typically pre-configured systems, ideal for users who want a reliable, ready-to-use computer with minimal customisation options.

Pre-built PCs offer a good balance of convenience and customisation, and they are a great choice for gamers, content creators, and professionals who don't want to build the system themselves but still want a specialised setup.

Building a PC gives the most flexibility, allowing you to select every component, which can result in a more tailored and cost-effective system, though it requires more time, effort, and knowledge.

Ultimately, the choice between these options depends on your specific needs, budget, and how hands-on you want to be in the process.

Operating systems



Operating systems

An Operating System (OS) serves as the foundational software layer that enables a computer to function efficiently.

It is responsible for managing the hardware components of the system, coordinating the utilisation of resources like the CPU, memory, storage, and input/output devices, and ensuring these components work together seamlessly.

Operating systems

At its core, an operating system provides a critical bridge between the hardware, software, and the user.

It acts as an intermediary that facilitates communication and interaction, enabling users and applications to make use of the underlying hardware without needing to understand the technical intricacies.

Without an operating system, a computer would be unable to perform even the most basic tasks, making the OS the most essential component of the computing environment.

Operating systems: Types

Real-time

- Objective of a real-time operating system is the quick and predictable response to events.
- Real-time operating systems have an event-driven or time-sharing design and often aspects of both.

Multi-user

- A multi-user operating system allows multiple users to access a computer system concurrently.

Distributed

- A distributed operating system manages a group of independent computers and makes them appear to be a single computer.

Operating systems: Types

Embedded

- Designed to operate on small machines like PDAs with less autonomy.

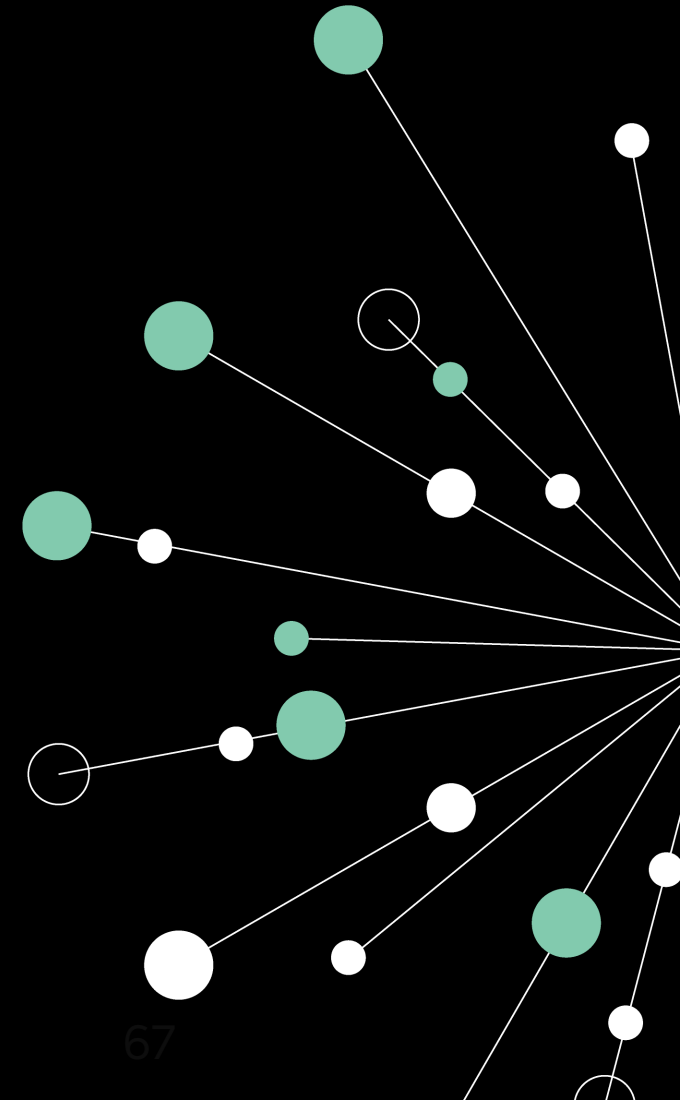
Multi-tasking and single-tasking

- When the operating system allows the execution of multiple tasks at one time, it is classified as a multi-tasking operating system.

Activity 11: OS types Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

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Operating systems: The kernel

In the context of operating systems, **the kernel** is the core component and central module that serves as the bridge between a computer's hardware and its software.

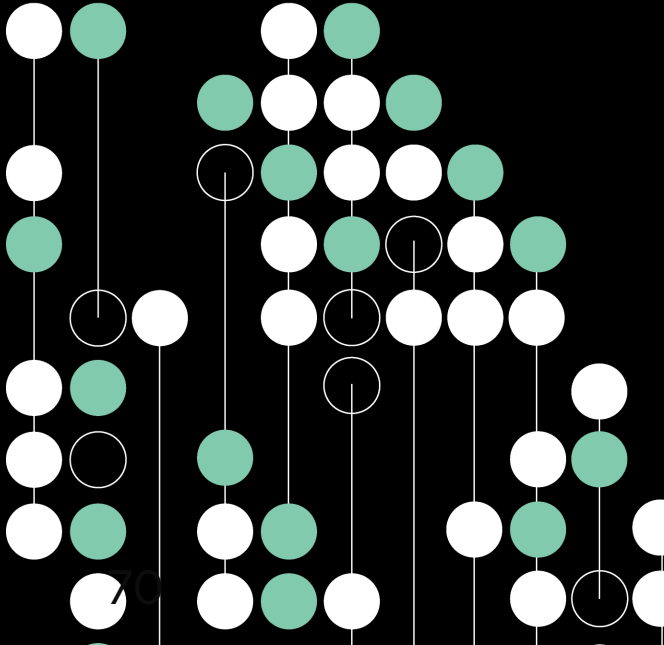
It is responsible for managing system resources, ensuring efficient communication between hardware devices and software applications, and maintaining overall system stability and security.

Operating systems: Common OS products

- Unix
- Linux
- Windows
- Apple
- Android

Unix

QA



Operating systems: UNIX

Unix is a family of multitasking, multiuser computer operating systems that derive from the original AT&T Unix, developed starting in the 1970s at the Bell Labs research centre by Ken Thompson, Dennis Ritchie, and others.

One of the key advantages of rewriting UNIX in C was its ability to run on a wide range of hardware platforms. Previously, operating systems were often tied to the specific hardware for which they were written, but UNIX's transition to C allowed it to transcend these limitations.

Distro's include Solaris, SCO UNIX and FreeBSD.

Many Unix-like operating systems have arisen over the years, of which Linux is the most popular, having displaced SUS-certified Unix on many server platforms since its inception in the early 1990s. Android, the most widely used mobile operating system in the world, is in turn based on Linux.

Operating systems: UNIX

A distribution operating system, commonly referred to as a "distro," is a version of an operating system that includes the core components (such as the kernel) along with additional software, tools, and configurations tailored for specific purposes or audiences.

Distributions are particularly common in the Linux and UNIX ecosystems, where the operating system's modularity allows for significant customisation.

While the term "operating system" refers to the core software that manages hardware and software resources (e.g., the Linux kernel or FreeBSD kernel), a distribution includes **the OS core plus additional software and customisations**.

It's like the difference between a car's engine (OS) and a fully assembled car (distro).

Operating systems: UNIX

UNIX is primarily used in industries and enterprise environments rather than by individual users or consumers. Its design, stability, and robustness make it well-suited for large-scale, mission-critical applications. Here's a detailed look at why and where UNIX is used in industry:

UNIX systems are widely used to power **enterprise servers and data centres**, where reliability and uptime are critical.

Telecommunications companies rely on UNIX for its advanced networking capabilities and scalability.

Banks and financial institutions use UNIX for tasks that require high security, stability, and performance, such as handling millions of transactions daily and managing real-time trading systems.

Operating systems: UNIX

FreeBSD serves as the foundation for the operating systems used in various PlayStation consoles. Sony has adapted FreeBSD for its gaming consoles due to its flexibility, stability, and permissive licensing. Here's a breakdown:

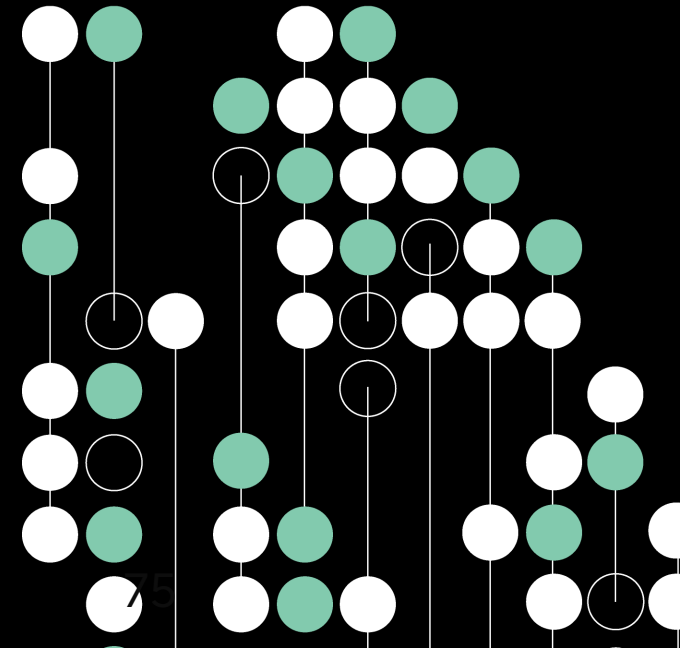
While the PS3 didn't use FreeBSD directly, it laid the groundwork for future consoles by incorporating elements of open-source software.

The PS4 operating system, called Orbis OS, is heavily based on FreeBSD 9.0. Sony modified and extended the OS to meet the specific needs of a gaming console.

The PS5 operating system is an evolution of Orbis OS, continuing to build on the FreeBSD foundation. Sony has further customised it to support the advanced features of the PS5.

Linux

QA



Operating systems: Linux

Linux is a Unix-like computer operating system assembled under the model of free and open-source software development and distribution.

The defining component of Linux is the Linux kernel, an operating system kernel first released on September 17, 1991, by Linus Torvalds.

The Free Software Foundation uses the name GNU/Linux to describe the operating system, which has led to some controversy.

Operating systems: Linux

Below are examples of where Linux is commonly utilised.

Servers and data centres

- Web servers
- Database servers
- Cloud computing
- Enterprise servers

Embedded systems

- Smart TVs and set-top boxes (e.g., Android TV, Roku).
- Networking devices like routers, switches, and firewalls.
- Automotive systems (e.g., Tesla's infotainment systems).
- Internet of Things (IoT) devices.

Operating systems: Linux

Linux is released under open-source licenses, meaning anyone can access, modify, and distribute its source code.

Most Linux distributions are available free of charge, reducing the total cost of ownership, especially for businesses and educational institutions.

Linux has strong security features, including user privilege management and a robust file permission system.

It is less prone to malware and viruses compared to other operating systems like Windows.

Linux can run on low-resource hardware, making it ideal for older systems or devices with limited processing power.

It is known for being efficient in terms of CPU and memory usage.

Operating systems: Linux

Linux-based distributions:

- **Ubuntu:** Popular desktop and server OS with a focus on ease of use.
- **Red Hat Enterprise Linux (RHEL):** A commercial OS for enterprise environments.
- **Debian:** Known for its stability and robustness; a base for many other distros.
- **Arch Linux:** A minimalist, rolling-release distro for advanced users.
- **Kali Linux:** Designed for penetration testing and security research.

Operating systems: Linux

Native Linux applications refer to software that is designed and optimised specifically to run on Linux systems, taking full advantage of the platform's features, file systems, and security mechanisms.

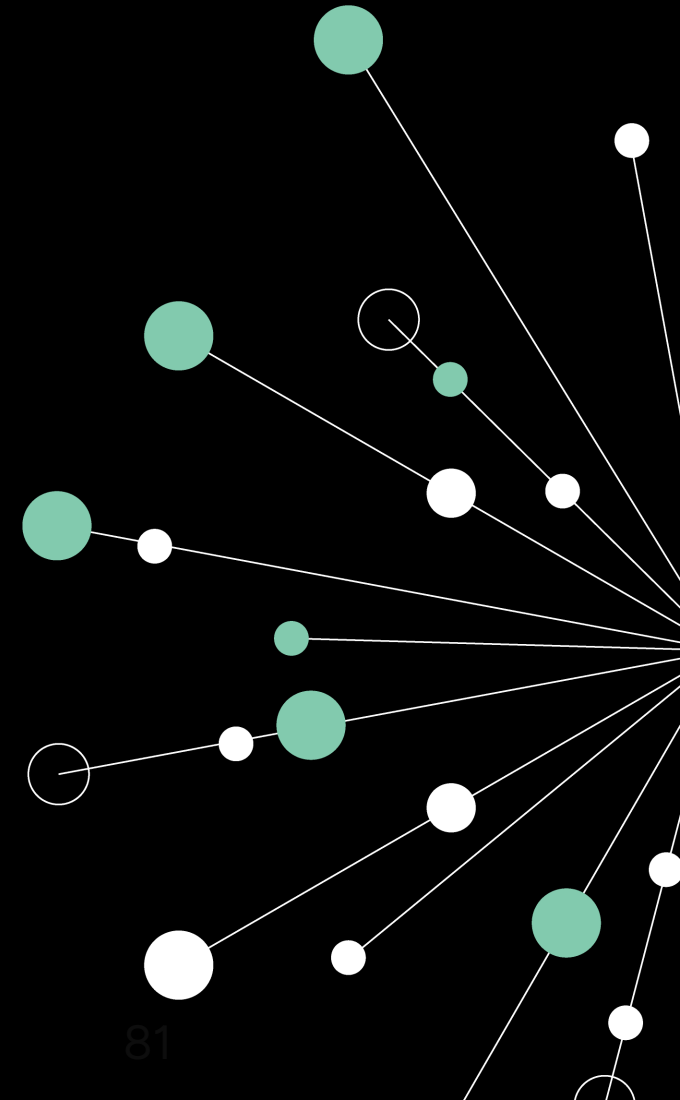
Here's a list of popular native Linux applications:

- **Firefox**
- **LibreOffice**
- **GIMP**
- **Audacity**
- **GNOME Terminal**
- **Nano / Pico**

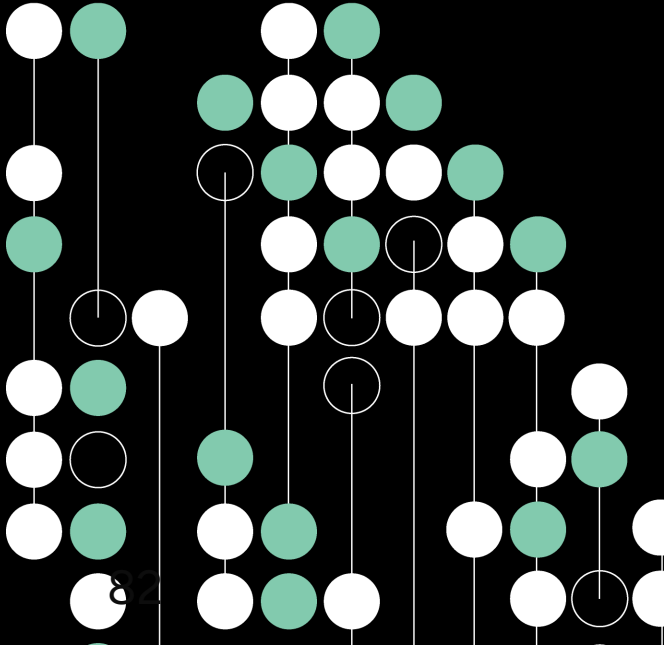
Activity 12: Linux Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Windows



Operating systems: Windows

MS-DOS (Microsoft Disk Operating System): Before Windows, Microsoft's main operating system was MS-DOS, a command-line-based OS that became widely used in the early 1980s. MS-DOS was essential to personal computing during this era, providing the basic interface for users to interact with the machine.

Microsoft officially launched Windows 1.0 on November 20, 1985. It was not a full operating system but an interface running on top of MS-DOS. It featured basic graphical elements like **windows, icons, menus, and pointers**, but was relatively limited in functionality. Windows 1.0 also lacked overlapping windows and multitasking, making it feel more like a GUI shell than a true OS.

Operating systems: Windows

In human-computer interaction, **WIMP** stands for 'windows, icons, menus, pointer', denoting a style of interaction using these elements of the user interface. It was coined by Merzouga Wilberts in 1980.

Though the term has fallen into disuse, some use it as an approximate synonym for Graphical User Interface (**GUI**). Any interface that uses graphics can be called a GUI, and WIMP systems derive from such systems.

However, while all **WIMP** systems use graphics as a key element (the icon and pointer elements), and therefore are **GUIs**, the reverse is not true. Some GUIs are not based in windows, icons, menus, and pointers.

For example, most mobile phones represent actions as icons, and some may have menus, but very few include a pointer or run programs in a window.

Operating systems: Windows

Below is an overview of the key uses of Windows OS across different sectors:

Home use: Windows is the most common operating system for personal computers worldwide. Users rely on Windows for everyday tasks such as web browsing, word processing, multimedia consumption, gaming, and social networking.

Corporate environments: Windows operating systems are used in businesses of all sizes due to their compatibility with enterprise applications, network services, and security features.

Virtualisation: Windows is widely used for running Virtual Machines (VMs) for various purposes such as testing software, creating isolated environments, and running legacy applications.

Operating systems: Windows

Windows does not use distributions (distros) in the same way that Linux does. The term 'distro' is specifically used in the context of Linux-based operating systems, referring to different variations or flavours of Linux.

Windows has different **editions** rather than distributions. These editions are tailored for specific use cases, hardware, and user needs these include:

Windows Home: Aimed at general home users with basic features.

Windows Pro: Designed for professionals with added features like BitLocker encryption, group policy management, and remote desktop.

Windows Enterprise: Targeted at large organisations with additional security and management features.

Windows Server Editions: Designed specifically for server environments (Windows Server 2019, 2022, etc.).

Operating systems: Windows

Windows **versions**, refers to the specific releases or builds of the Windows operating system over time.

A version generally refers to a major update or iteration of the Windows OS, which could include changes, new features, improvements, or a complete overhaul of the system.

Examples:

- Windows 11
- Windows 10
- Windows Vista
- Windows XP
- Windows 95
- Windows Server 2019

Operating systems: Windows

Here is a list of some of the **native apps** (built-in applications) that come with Windows operating systems.

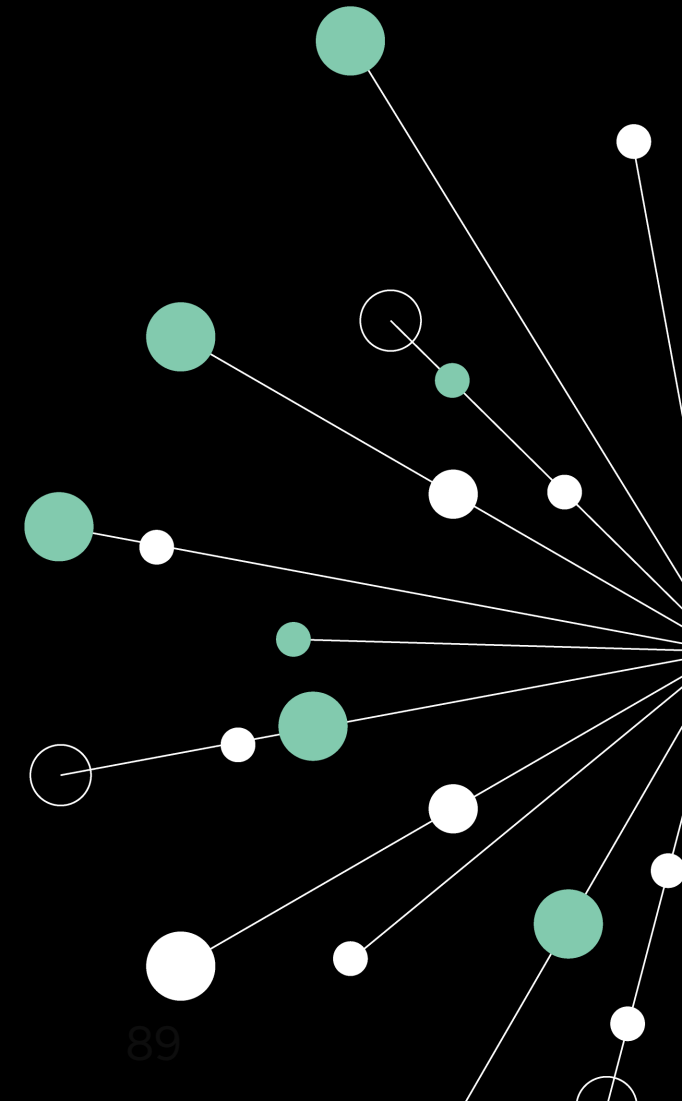
These apps are pre-installed with Windows, and their availability may vary depending on the Windows version (e.g., Windows 10, Windows 11):

- Cortana
- Sticky Notes
- Windows Security
- Xbox Game Bar
- Paint
- Photos
- Microsoft Store

Activity 13: Windows Q&A

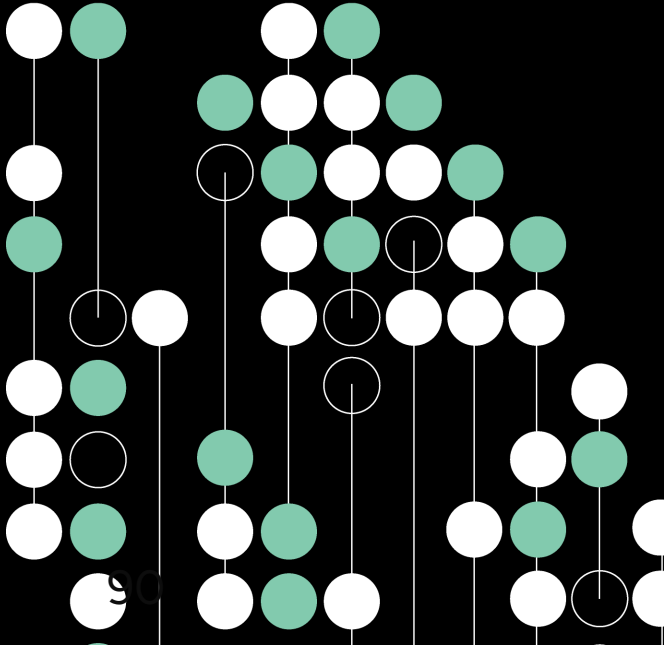
Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Apple

QA



Operating systems: Apple

The history of **Apple's operating systems** is a rich and evolving journey that spans several decades, with significant milestones in both desktop and mobile computing.

Apple's operating systems have undergone numerous transformations, from their early days to their current modern iterations.

Apple's journey into operating systems began with the release of **System Software 1.0** for the Macintosh in 1984.

This early OS was designed to be simple and user-friendly, with a Graphical User Interface (GUI) that revolutionised the personal computing industry.

The Macintosh operating system was built on a proprietary kernel and was a significant departure from the text-based systems like MS-DOS, which dominated the market at the time.

Operating systems: Apple

Throughout the late 1980s and early 1990s, Apple continued to refine its operating system with successive updates, adding features like virtual memory, improved file systems, and better hardware support. By the early 1990s, the system was rebranded as **Mac OS**.

The turning point for Apple OS came in 2001, when Apple introduced **Mac OS X**, which was a completely reworked version of its **Mac OS**. **Mac OS X** was built on the Unix-based NeXTSTEP operating system.

Over the next decade, Apple released several **iterations** of **Mac OS X**, each improving performance, stability, and adding new features

In 2012, Apple rebranded **Mac OS X** to **OS X**, signalling a more streamlined and modern approach.

In 2016, Apple rebranded **OS X** to **macOS**, aligning the naming convention with its mobile operating systems (iOS, watchOS, and tvOS).

Operating systems: Apple

Apple uses the term **versions** to describe the different releases and updates of its operating systems.

Apple's operating systems are released in **numbered versions**, typically tied to major updates. For example: macOS (formerly OS X) has versions like macOS 10.15 (Catalina), macOS 11 (Big Sur), macOS 12 (Monterey), and macOS 13 (Ventura).

iOS also follows a versioning system, with updates like iOS 14, iOS 15, and iOS 16, etc.

Each version introduces new features, security improvements, and bug fixes. The major version changes are **usually named** (e.g., **Big Sur** for macOS 11 or **Monterey** for macOS 12), while **point releases** (e.g., 10.15.1 or 16.1) usually focus on incremental updates and patches.

Operating systems: Apple

Apple also refers to certain versions as **editions**, especially for specialised hardware, but this term is not as commonly used as versions.

For example, **macOS Server** (the server-oriented version of macOS) can be considered an edition of macOS, designed for more business or enterprise-focused needs.

Operating systems: Apple

Apple provides a suite of native apps that come pre-installed on its devices, including macOS, iOS, iPadOS, watchOS, and tvOS.

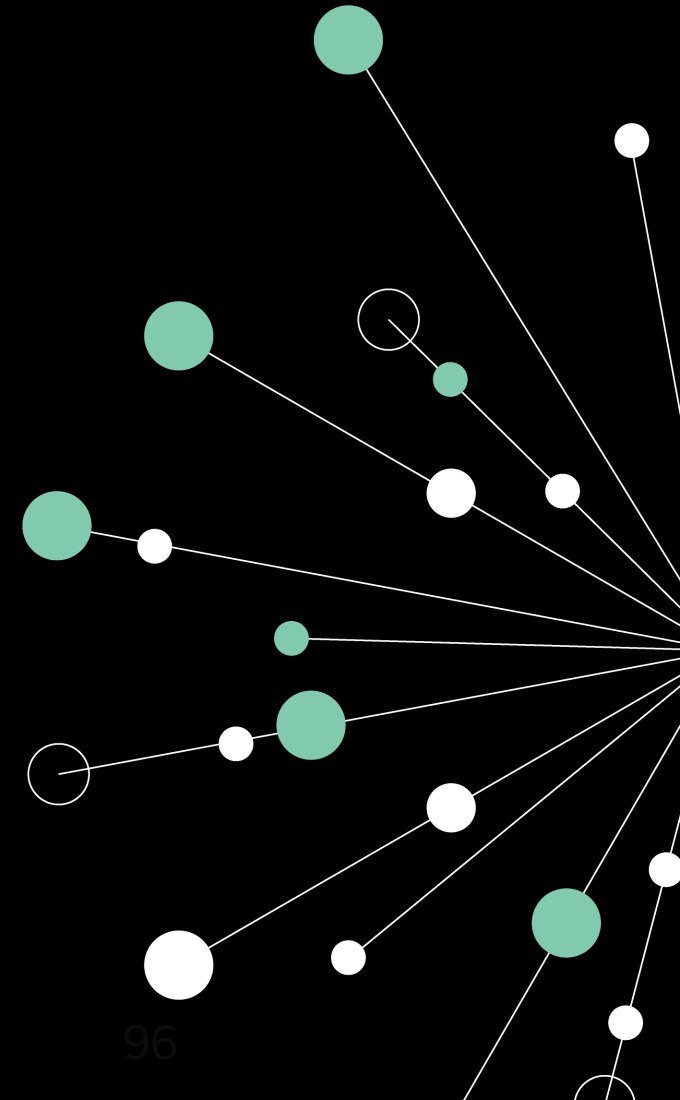
These apps are deeply integrated into the ecosystem and provide a wide range of functionality, from communication to entertainment and productivity. Examples include:

- App Store
- iCloud Drive
- Siri
- Maps
- FaceTime
- Safari

Activity 14: Apple Q&A

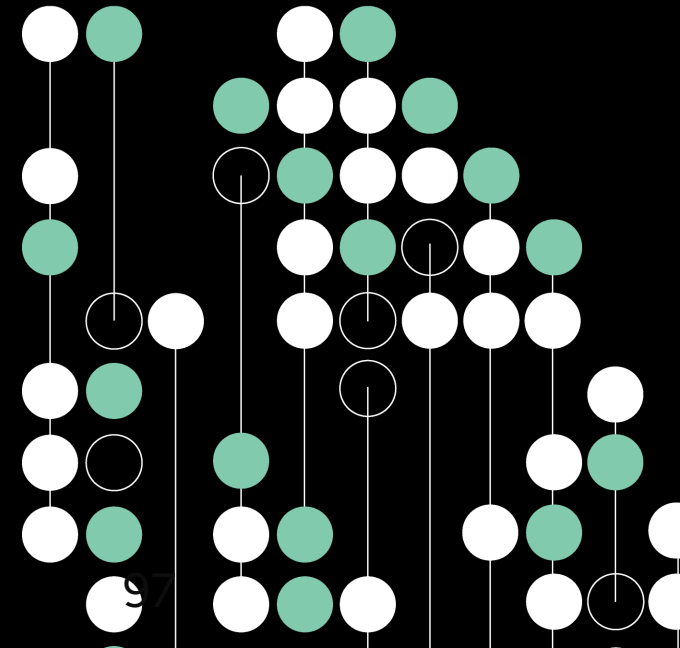
Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Android

QA



Operating systems: Android

In 2003 Android Inc. was founded in Palo Alto, California, by Andy Rubin, Rich Miner, Nick Sears, and Chris White. Initially, the company aimed to develop an advanced operating system for digital cameras.

The focus soon shifted to creating a platform for mobile devices due to the broader market potential.

Then in 2005 Google acquired Android Inc. for approximately \$50 million. Andy Rubin, along with his team, joined Google.

In 2008 The first public release of Android was **Android 1.0**, accompanied by the launch of the first Android smartphone, the HTC Dream (also known as T-Mobile G1).

Key features included an early version of the Android Market (now Google Play Store), a web browser, and basic apps like Gmail and Google Maps.

Operating systems: Android

Google adopted a tradition of naming Android versions after desserts in alphabetical order:

- **Cupcake (1.5):** Introduced in 2009, featuring an on-screen keyboard and widgets.
- **Donut (1.6):** Brought CDMA network support, expanding Android's reach to more carriers.
- **Eclair (2.0–2.1):** Added features like live wallpapers, voice-guided navigation, and HTML5 support in the browser.
- **Froyo (2.2):** Boosted performance, introduced the portable hotspot feature, and improved app functionality.

Operating systems: Android

Android has versions and editions that cater to different devices, user needs, and use cases.

Android **versions** are the updates to the core operating system, bringing new features, performance improvements, and security enhancements.

Versions are numbered (e.g., Android 11, 12, 13).

Up to Android 9, versions were named after desserts in alphabetical order (e.g., Lollipop, Marshmallow, Pie).

Starting with Android 10, the dessert naming convention was dropped in favour of a simpler numerical approach.

Operating systems: Android

Android has several specialised **editions** tailored for specific device types and use cases:

- **Standard Android:** Android without any additional modifications or customisations by manufacturers.
- **Android One:** Android designed for mid-range and budget devices, ensuring a clean and efficient experience.
- **Android Go Edition:** A lightweight edition of Android designed for entry-level devices with limited hardware resources (low RAM and storage).
- **Android TV:** Optimised for televisions like Sony, TCL, and Hisense.
- **Android Automotive:** Tailored for in-vehicle infotainment systems like Volvo, Polestar, and General Motors.

Operating systems: Android

Android native apps are applications developed specifically to run on Android devices, leveraging the Android SDK (Software Development Kit) and taking full advantage of the platform's features.

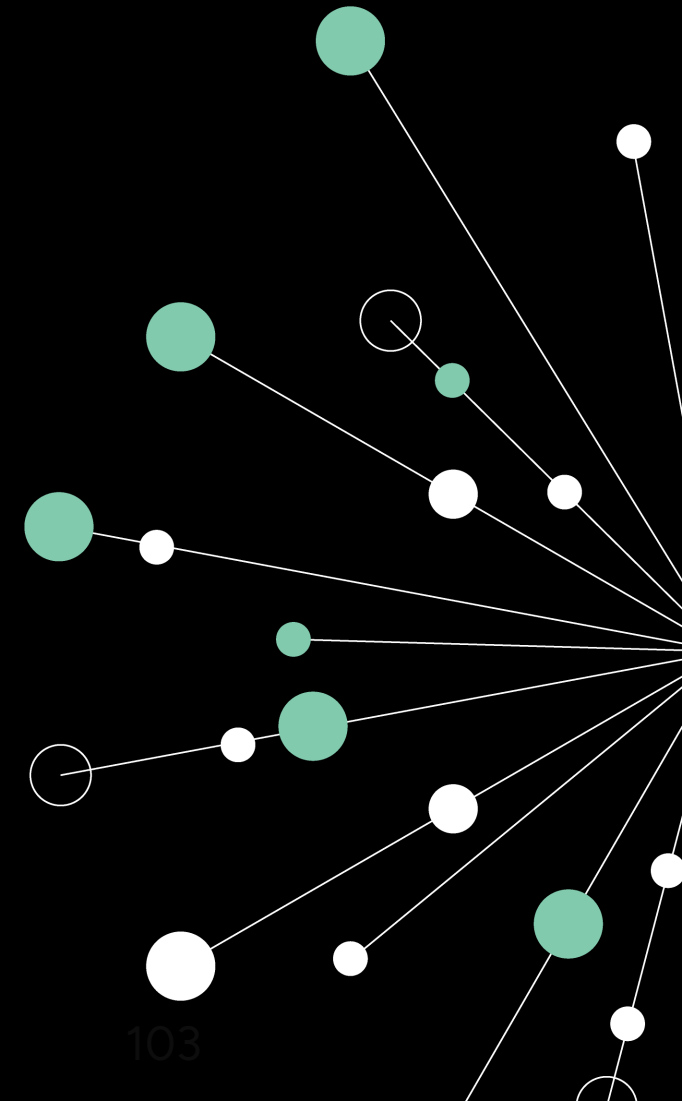
Here's a list of native Android apps:

- Google Find My Device
- Phone/Dialler
- Google Duo
- YouTube
- Google Map
- Google Chrome
- Google Play Store

Activity 15: Android Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: Operating systems

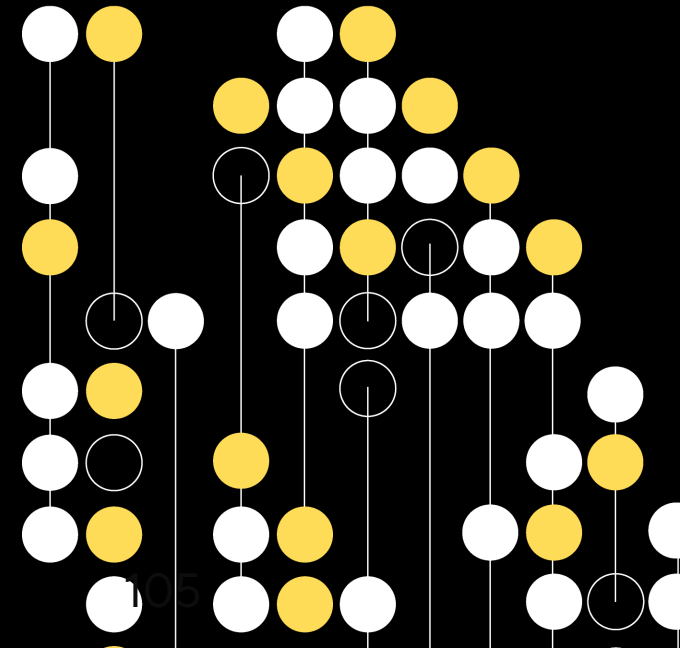
Operating Systems (OS) form the backbone of modern computing, enabling devices to run applications, manage resources, and interact with users.

Over the years, major operating systems have developed various versions and editions tailored to diverse use cases, from personal computing to enterprise environments.

From user-friendly platforms like Windows and macOS to the adaptability of Linux and the mobility focus of Android and iOS, each OS brings unique strengths to the computing ecosystem.

Their continuous evolution ensures they remain relevant, secure, and capable of meeting the demands of an ever-changing technological world.

Operating system install



Operating system install: Pre-installation planning

Proper planning is essential before installing an Operating System (OS), especially for servers, to ensure it meets organisational requirements and operates efficiently.

Careful pre-installation planning ensures that the chosen operating system aligns with organisational goals, budget, and performance needs.

By addressing server role, budget, system specifications, performance requirements, fault tolerance, and stakeholder coordination, you can achieve a smooth and efficient deployment that supports long-term success.

Operating system install: Pre-installation planning

Environment: Assess the physical and virtual environments where the OS will be installed, such as a desktop workstation, server, or virtual machine.

Physical security: Ensure the system is in a secure area to prevent unauthorised physical access.

Cooling requirements: Evaluate the cooling requirements of the hardware where the OS will run, especially for high-performance systems like servers.

Network access: Verify that the system has reliable access to the network for installation and post-installation updates.

Protocol support: Verify that the system has reliable access to the network for installation and post-installation updates.

Client support: Ensure the OS supports the devices it will interact with, such as printers, scanners, or external peripherals.

Operating system install: Pre-installation planning

There are several installation methods available depending on the system's infrastructure, the scale of deployment, and the specific requirements of the environment:

DVD installation: OS installation files are burned onto a bootable DVD.

USB installation: A bootable USB drive containing the OS installation files.

Unattended installation: Automated installation process that eliminates the need for manual input during setup.

Use of an answer file: Unattended installation using an answer file (a pre-configured XML or text file containing responses to installation prompts (e.g., language settings, partitioning, product keys)).

Operating system install: Pre-installation planning

Methods which remove the need for physical media (for environments where the OS is installed over a network):

Network share containing setup files: The OS installation files are hosted on a network share, accessible to the target system.

Network booting (PXE – Preboot Execution Environment): Allows systems to boot and install an OS directly over the network without local media.

TFTP (Trivial File Transfer Protocol): A lightweight protocol often used in conjunction with PXE for transferring boot images and installation files to the target system.

Operating system install: Pre-installation planning

Image-based installation is a highly efficient way of deploying operating systems across multiple machines or environments.

This method involves creating an exact replica (or image) of an OS setup, which can be deployed to new systems quickly.

It saves time, ensures consistency, and simplifies the deployment of large-scale IT infrastructures.

Operating system install: Pre-installation planning

Windows SIM is part of the **Windows Assessment and Deployment Kit (ADK)** and is used to create unattended Windows installation answer files (also known as XML files).

Drive cloning involves making a direct copy of a disk or partition, including the OS and all installed applications, files, and settings.

Sysprep is a tool provided by Microsoft to prepare an OS for disk imaging or duplication.

WDS is a Microsoft server role that provides the infrastructure to deploy Windows OS over the network.

MDT is a solution accelerator from Microsoft that helps IT professionals automate the deployment of Windows operating systems.

Operating system install: Pre-installation planning

Third-party tools provide advanced features for OS deployment, such as backup, recovery, and cloning capabilities:

Symantec Ghost (formerly known as Norton Ghost) is a disk cloning and imaging tool. Allows users to capture an OS image and deploy it across multiple systems.

Altiris provides automated deployment of OS images and software across large networks.

Macrium Reflect is a powerful disk imaging and cloning tool, known for its reliability and efficiency. It allows users to create full disk images or incremental backups of the OS and other data.

Operating system install: Starting the OS

When you power on a Personal Computer (PC), several critical processes occur to initialise hardware and prepare the system for use.

These processes are part of the boot sequence and involve **POST**, **CMOS**, and **BIOS**.

Operating system install: Starting the OS

POST is a diagnostic process that occurs when the computer is powered on. It checks the basic functionality of the hardware components to ensure everything is working correctly.

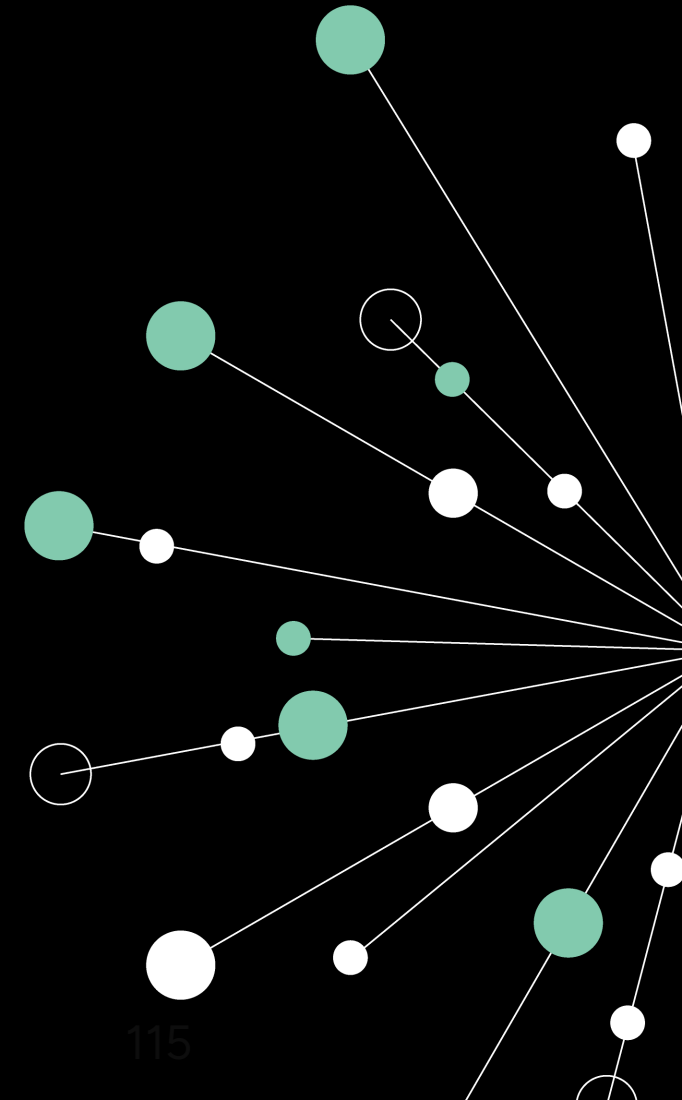
CMOS refers to the type of memory used to store configuration settings for the system, such as the time, date, and system configuration data (e.g., boot device order, hardware settings).

BIOS is firmware embedded on the motherboard that initialises and tests hardware components during boot and provides an interface between the operating system and the hardware.

Activity 16: BIOS vs UEFI Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Operating system install: UEFI vs BIOS

UEFI offers significant improvements over BIOS, such as support for larger drives, more partitions, a modern graphical interface, and enhanced security features like secure boot.

The backing from major tech organisations ensures that UEFI remains a key part of the modern computing landscape, offering flexibility, scalability, and security.

Unlike BIOS, which is limited to four primary partitions on a hard drive (HDD), UEFI uses the GUID Partition Table (GPT), which allows for 128 partitions on a single drive.

Operating system install: UEFI vs BIOS

BIOS uses the Master Boot Record (MBR) partitioning scheme, which supports disks up to 2TB. In contrast, UEFI, utilising GPT, supports disks much larger than 2TB, enabling the use of drives in the petabyte and exabyte range.

UEFI, through GPT, has a theoretical limit of about 9.4 Zettabytes (9.4 billion terabytes) for partition sizes. This is far beyond current technological needs, ensuring that UEFI is future-proof and capable of handling much larger storage capacities as technology advances.

While UEFI introduces advanced features, it also includes a legacy BIOS compatibility mode (often called CSM for Compatibility Support Module) that allows the system to boot older operating systems or software designed for BIOS-based systems.

Operating system install: UEFI vs BIOS

UEFI is widely supported and promoted by major companies in the technology industry:

- **IBM:** Contributes to UEFI's development, particularly for enterprise and server solutions.
- **Intel:** Has actively supported UEFI in its chipset designs, improving the efficiency of boot processes.
- **Microsoft:** UEFI is required for the secure boot process in modern versions of Windows, enhancing system security.
- **AMD:** Ensures compatibility with UEFI in its processors and motherboards, making it a standard for modern computing hardware.
- **Apple:** macOS leverages UEFI in its hardware, ensuring faster booting and security for its computers.

Operating system install: Out-of-Box Experience (OOBE)

Out-of-Box Experience (OOBE) refers to the initial setup process that a user goes through when they first power on a new device, such as a computer, smartphone, or software.

It's the very first interaction a user has with a device after it is unboxed and powered on.

The OOBE is designed to guide the user through configuring the system for their personal use, and it typically involves a series of steps that help them set up essential features and preferences.

A well-designed OOBE process helps users get the most out of their new device with minimal effort.

Operating system install: Post-installation configuration

Once the OS is installed on a server, a variety of post-installation configuration tasks need to be completed to ensure that the server is functional, secure, and ready for use.

These tasks include server configuration, network configuration, applying security settings, and setting up required services.

Operating system install: Post-installation configuration

- Enable Remote Desktop
- Set time zone
- Change computer name, add domain suffix or
- Join the domain
- Network configuration
- Shared resources
- Power settings
- Security settings
- Firewall
- User accounts
- Disk space used to provide secondary RAM
- Automatic Updates
- WSUS

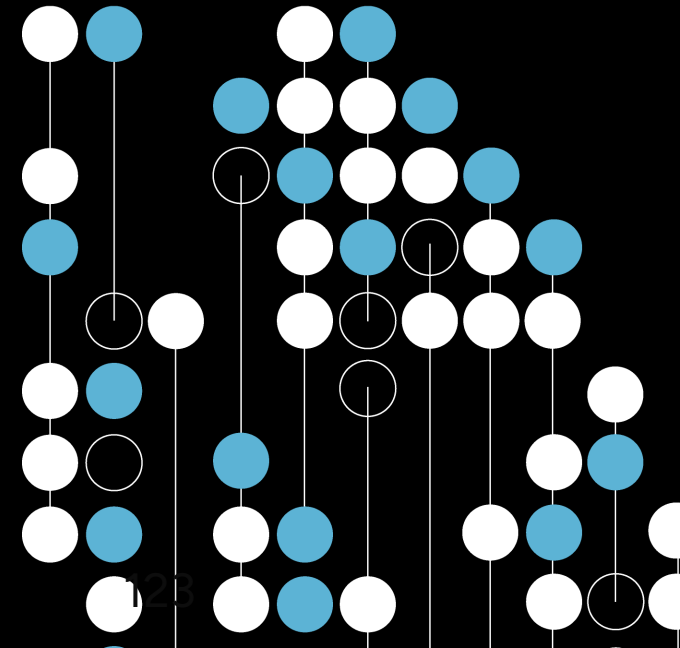
In summary: Operating system install

Thorough pre-installation planning ensures that all necessary configurations, hardware components, and security measures are in place.

This careful preparation is crucial to prevent any disruptions, minimise potential compatibility issues, and ensure the OS operates efficiently after installation.

Proper planning and attention to detail during this phase provide the foundation for a secure, stable, and effective system.

Principles of virtualisation



Principles of virtualisation

Virtualisation is a software-based solution that is fundamentally changing the way many companies utilise their hardware resources.

Modern computer hardware was originally designed to run a single operating system and, in some cases, for a variety of reasons including increased security, a single application.

However, virtualisation technology makes it possible to simultaneously run multiple operating systems and multiple applications on the same computer.

In essence, virtualisation lets you transform hardware into software.

Principles of virtualisation

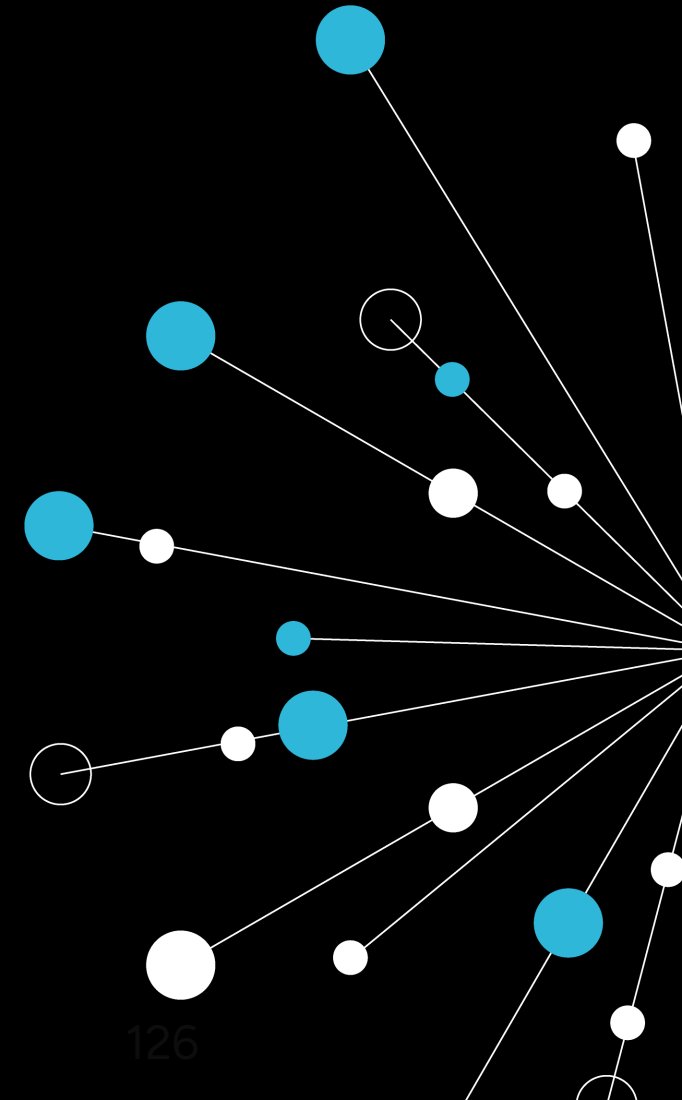
It improves resource utilisation, simplifies management, and reduces costs. Here are the main forms of virtualisations:

- Cloud virtualisation
- Data virtualisation
- Application virtualisation
- Desktop virtualisation
- GPU virtualisation
- Network virtualisation
- Storage virtualisation
- Hardware virtualisation
- Software virtualisation

Activity 17: Virtualisation Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Principles of virtualisation

Virtual Machines (VMs) offer a versatile and efficient solution for a wide range of IT and development needs.

By virtualising the hardware and providing an isolated operating environment, VMs enhance flexibility, security, and productivity.

Principles of virtualisation

Key reasons for using VMs include:

- Provides separation from the kernel and wider system
- Good for testing malware in a sanitised environment
- Great for developers and architects
- Easier to test interactions of distributed architectures
- Operate in a safe environment to boost confidence
- Snapshots save and revert to certain states with ease
- Easier to share work states

Principles of virtualisation

A hypervisor, also known as a Virtual Machine Monitor (VMM), is software or firmware that creates and manages Virtual Machines (VMs) by abstracting physical hardware resources.

There are two primary types of hypervisors:

1. Type 1 (Bare-Metal)
2. Type 2 (Hosted)

Principles of virtualisation

Type 1 hypervisors run directly on the host's hardware, without requiring a host operating system.

They are installed on the physical machine and manage VMs directly.

Key features:

- Direct access to hardware resources.
- High performance and efficiency.
- Minimal overhead since no intermediary OS is present.

Principles of virtualisation

Type 2 hypervisors run on top of a host operating system and rely on it for resource management.

They are software applications installed on an existing OS.

Key features:

- Easy to install and use.
- Ideal for smaller workloads or development purposes.
- Shares resources with the host OS, leading to some performance overhead.

Principles of virtualisation

Virtualisation enables users to access operating systems, desktops, and applications in a flexible and efficient manner by abstracting them from physical hardware.

Virtual desktops involve delivering a virtualised Operating System (OS) to users over the internet, allowing them to interact with a desktop environment that is hosted remotely.

Key features:

- Often delivered as part of Desktop as a Service (DaaS) solutions.
- The operating system appears as though it is running locally on the user's device.

Principles of virtualisation

Hosted remote desktops allow users to access a physical desktop system or virtual machine remotely over the internet.

Key features:

- The user interacts with a remote system via a client application or browser-based interface.
- The desktop environment is displayed in a window on the local device.

Hosted applications are specific software programs that users can access over the internet, typically through a virtualised environment.

Key features:

- The application runs on a remote server but appears and behaves as if it is installed locally on the user's device.
- Does not require local installation, simplifying deployment.

Principles of virtualisation

When setting up a Virtual Machine (VM), several factors must be carefully configured to ensure optimal performance and compatibility.

These considerations include:

Boot order: Define the sequence in which the VM checks devices.

Resource allocation: Determine the number of processor cores assigned to the VM to balance performance and host system capacity.

Memory (RAM): Allocate sufficient memory to support the guest OS and applications without overburdening the host.

Principles of virtualisation

Storage: Specify the storage size and format for the VM, ensuring enough space for the OS, applications, and data. Common formats include VDI, VMDK, and others.

Networking options: Configure the VM's network settings, such as bridged, NAT, or host-only, to ensure proper communication with other systems as required.

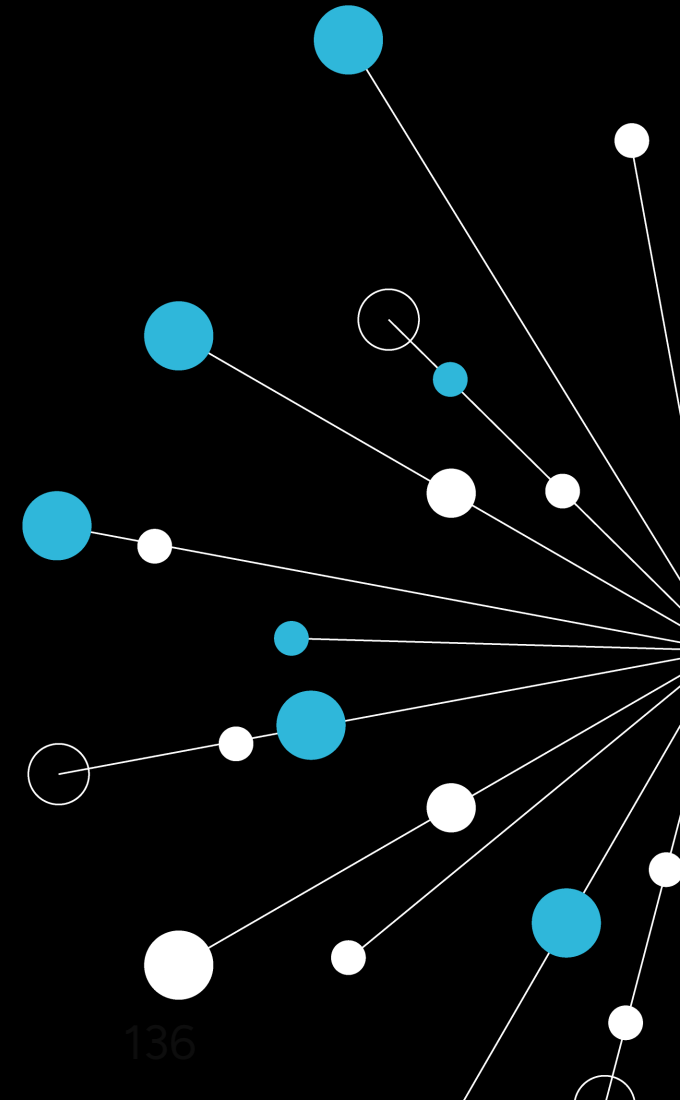
USB device access: Decide which USB devices will be accessible to the VM and in what format. For example, external drives or peripherals might need to be directly passed through.

Display settings: Set the VM's display resolution, number of monitors, and graphics acceleration to optimise the user experience.

Activity 18: Create a virtual machine

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

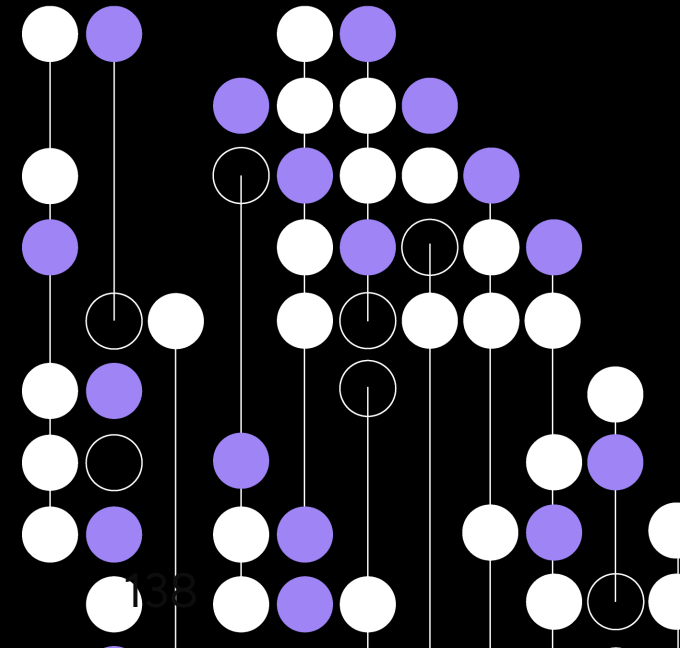
If you need any assistance just message the tutor.



In summary: Principles of virtualisation

Virtual Machines (VMs) have revolutionised the way we approach computing, offering flexibility, efficiency, and cost-effectiveness in diverse applications. They enable the creation of isolated environments for operating systems and applications, leveraging virtualisation to optimise resource utilisation and enhance security.

Operating System Security and Updates



Operating system security and updates

Operating System (OS) security and updates are critical for maintaining the integrity, reliability, and functionality of a computer system.

They ensure protection against emerging threats, enhance system performance, and deliver new features to users.

These include:

- user authentication.
- access control.
- data encryption.
- firewall protection.
- malware protection.
- security logging.

Operating system security

User authentication: Ensures only authorised users can access the system through credentials such as passwords, biometrics, or multi-factor authentication.

Access control: Limits user and application permissions based on predefined roles to prevent unauthorised actions.

Data encryption: Protects sensitive data by converting it into a secure format that unauthorised users cannot easily decipher.

Firewall protection: Monitors and controls incoming and outgoing network traffic to block unauthorised access.

Malware protection: Includes built-in or third-party tools to detect and mitigate viruses, worms, spyware, and ransomware.

Security logging: Tracks user activity, system events, and potential threats for analysis and incident response.

Operating system updates

OS updates are periodic releases of patches, enhancements, and new features to improve the performance, functionality, and security of an operating system:

- **Security patches:** Address vulnerabilities and protect against newly discovered threats.
- **Bug fixes:** Resolve issues that affect system stability or performance.
- **Feature updates:** Introduce new functionalities or enhance existing ones.
- **Driver updates:** Ensure compatibility and optimal performance with hardware components.

Operating system updates: Update mechanisms

Automatic updates: Enable the system to download and install updates without user intervention, ensuring timely application of critical patches.

Manual updates: Allow users or administrators to control when and how updates are installed.

Best practices for updates:

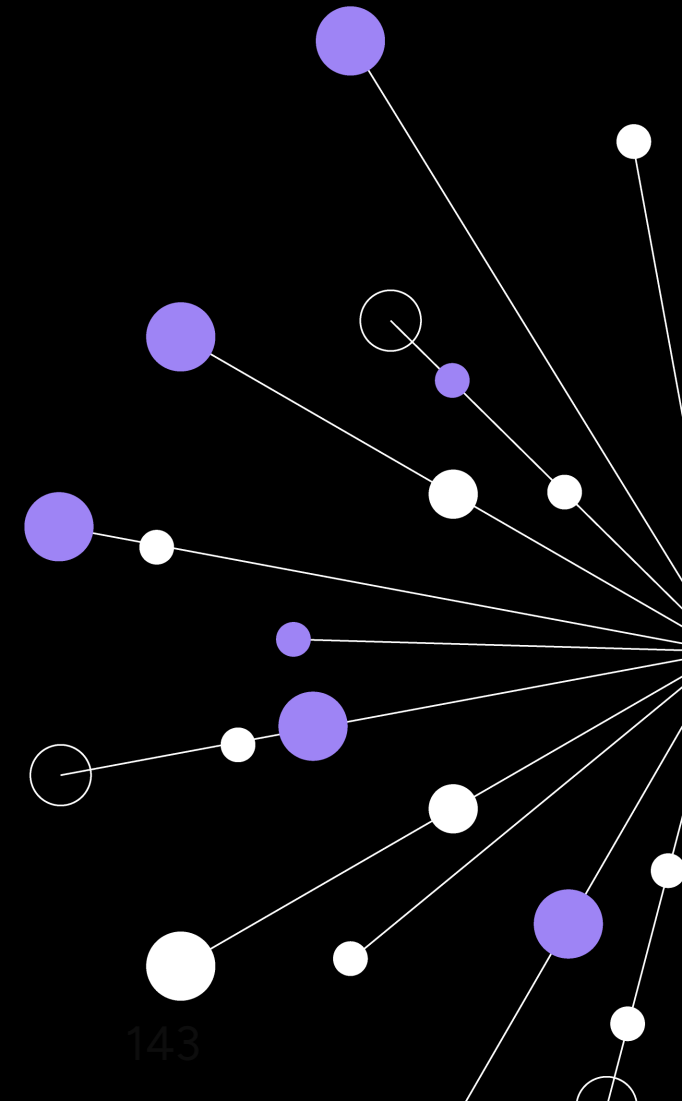
- Regularly check for updates and apply them promptly.
- Backup critical data before major updates to prevent data loss.
- Test updates in a non-production environment, particularly in enterprise settings.

Activity 19: OS security and updates

Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



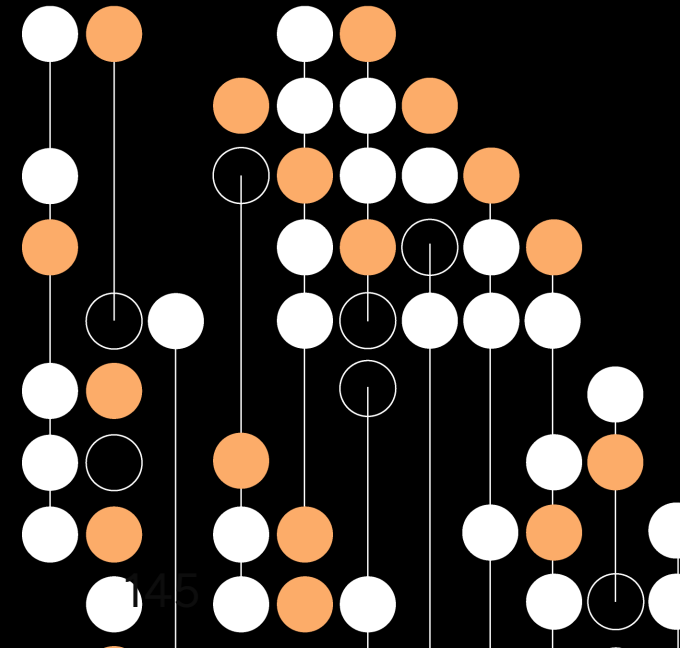
In summary: Operating system security and updates

Operating system security and updates are integral to safeguarding systems and ensuring seamless operation.

By implementing robust security measures and staying up-to-date with patches and enhancements, users and organisations can protect their systems from threats and enhance productivity.

- **Protection against cyber threats:** Timely updates mitigate vulnerabilities that hackers might exploit.
- **System stability and performance:** Updates resolve software bugs, ensuring a smoother user experience.
- **Regulatory compliance:** Staying updated helps organisations adhere to cybersecurity regulations and standards.
- **Extended system lifespan:** Proper security practices and updates reduce the risk of obsolescence and enhance the system's longevity.

System tools and resource management



System tools and resource management

In Windows, **system tools and resource management tools** are both critical for maintaining the operating system, but they serve different purposes and functionalities.

System tools and resource management

System tools:

- Provide tools for general system maintenance, configuration, and monitoring.
- Broad, covering maintenance, customisation, and performance.

Example: **Event viewer** to track system events and logs. It allows for basic control over system settings.

System tools and resource management

Resource management:

- Focus on efficient allocation, monitoring, and management of hardware and system resources.
- Specific to managing resources like CPU, memory, storage, and network.

Example: **Resource monitor** and **Task manager** show real-time resource usage. It allows fine-grained control over resource allocation and utilisation.

System tools and resource management: Examples

System tools focus on broader system management, while resource management deals with fine-tuning and customisation.

System tools	Resource management tools
Task scheduler	Automates tasks like backups or maintenance.
Control panel	Manages user accounts, system settings, and installed programs.
System restore	Reverts the system to a previous stable state.
Event viewer	Logs system and application events.
Defragment and optimise drives	Optimises storage but isn't focused on real-time resource monitoring.

System tools and resource management

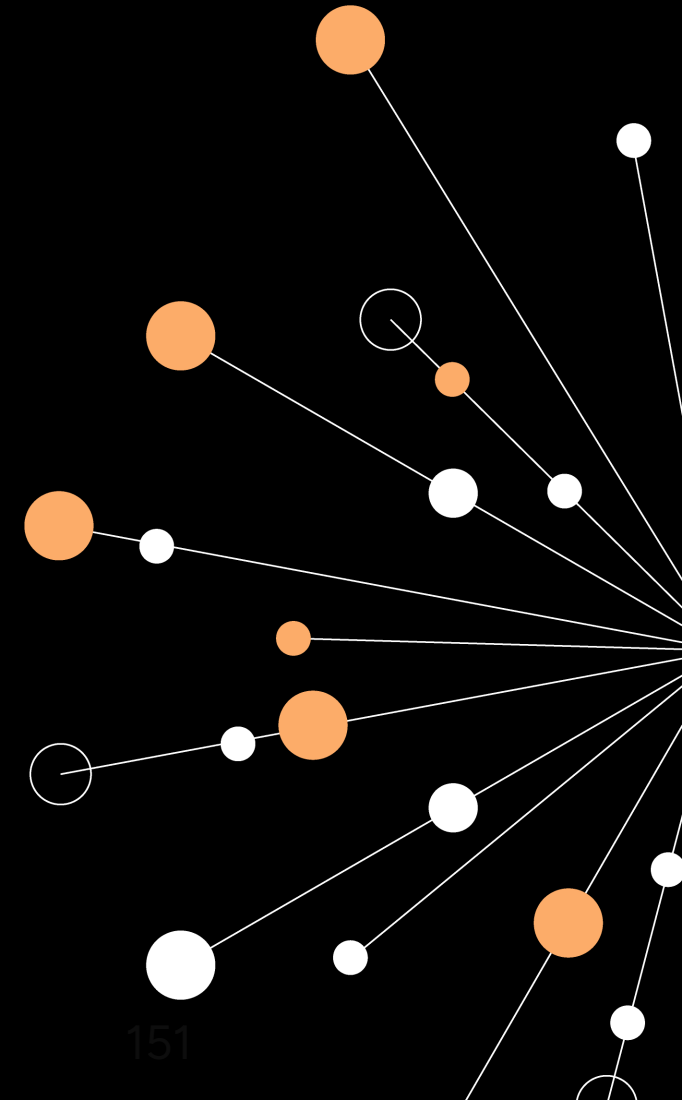
When performing diagnostics in the workplace, the breakdown highlights the importance of utilising both system tools and resource management.

Use case	System tools	Resource management
Troubleshooting	Resolve software issues or misconfigurations (e.g., using System restore).	Resolve performance bottlenecks or overutilisation issues (e.g., Task manager).
Performance tuning	Optimise system settings for overall stability.	Focused on allocating and balancing resource demands efficiently.
Maintenance	Schedule and perform tasks like updates, cleanup, or backups.	Regularly balance or free up resources for consistent performance.

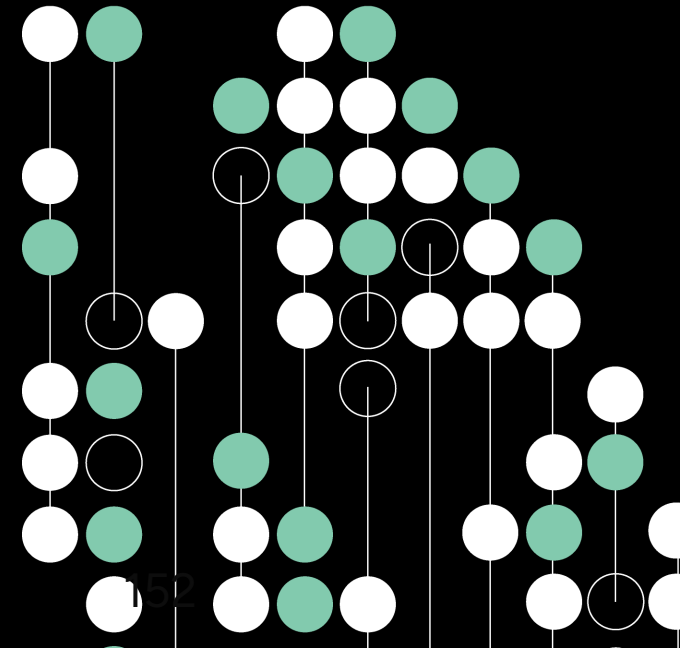
Activity 20: System tools and resource management Q&A

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Event viewer and log files



Event viewer and log files

Log files are records automatically generated by an operating system, applications, or services to capture events, errors, and system activity.

They are critical for monitoring, troubleshooting, and maintaining a secure and stable system.

Event viewer and log files

Event logging: Tracks actions such as logins, system errors, and application failures.

Debugging: Provides detailed error messages that developers and IT professionals can use to identify and fix problems.

Auditing: Logs user actions to ensure compliance with security policies.

Event viewer and log files

Types of Logs:

- **System Logs:** Contain operating system-related events (e.g., hardware failures, kernel errors).
- **Application Logs:** Record the activity of installed software.
- **Security Logs:** Track access attempts, successful logins, and potential security breaches.

Event viewer and log files

Location of log files:

Windows:

Most logs are accessible through the **Event viewer** (eventvwr), with files stored in '%SystemRoot%\System32\winevt\Logs'.

Linux:

Logs are stored in the /var/log/ directory.

Examples include: /var/log/syslog or /var/log/messages for system logs./var/log/auth.log for authentication and security logs.

Mac:

Logs can be found in the Console application or under /var/log/.

Event viewer and log files

Troubleshooting

Scenario:

- A computer crashes unexpectedly.

Action:

- Review the system log for error codes and descriptions to pinpoint the cause.

Security

Scenario:

- A suspicious login attempt is detected.

Action:

- Examine the security log to track the IP address and verify user credentials.

Event viewer and log files

Performance

Scenario:

- A server slows down under high load.

Action:

- Analyse performance logs to find resource bottlenecks.

Compliance

Scenario:

- An organisation needs to demonstrate compliance with data protection regulations.

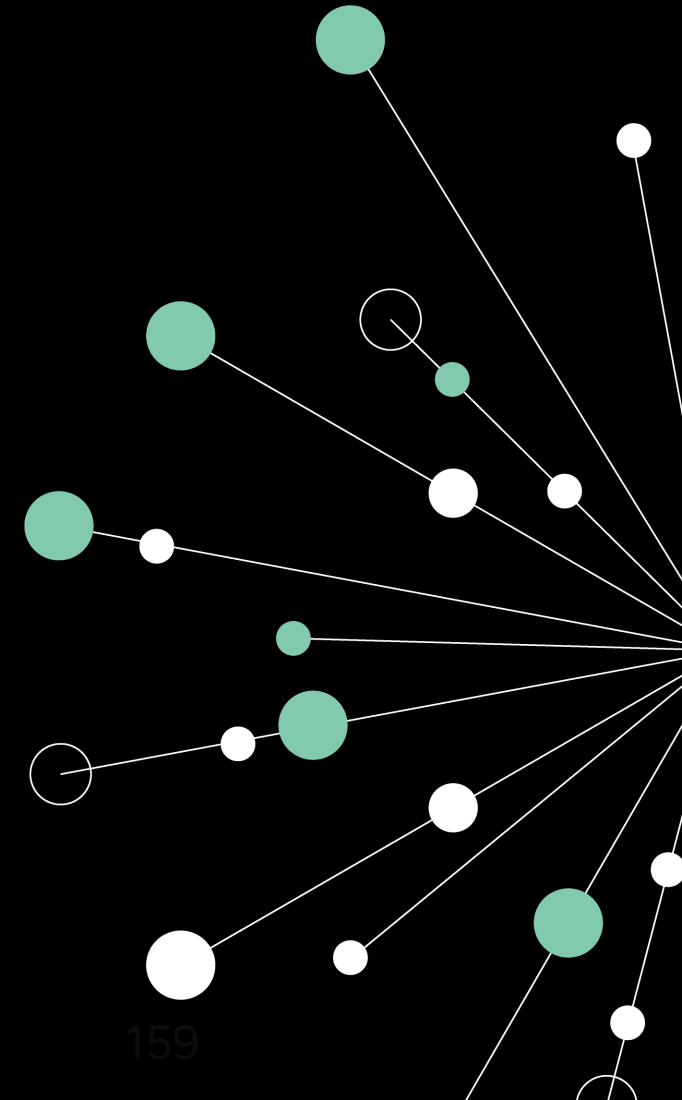
Action:

- Use logs to show records of system activity and user access.

Activity 21: Practical on event viewer and log files

Complete the activity in your Exercise Guide, then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: Event viewer and log files

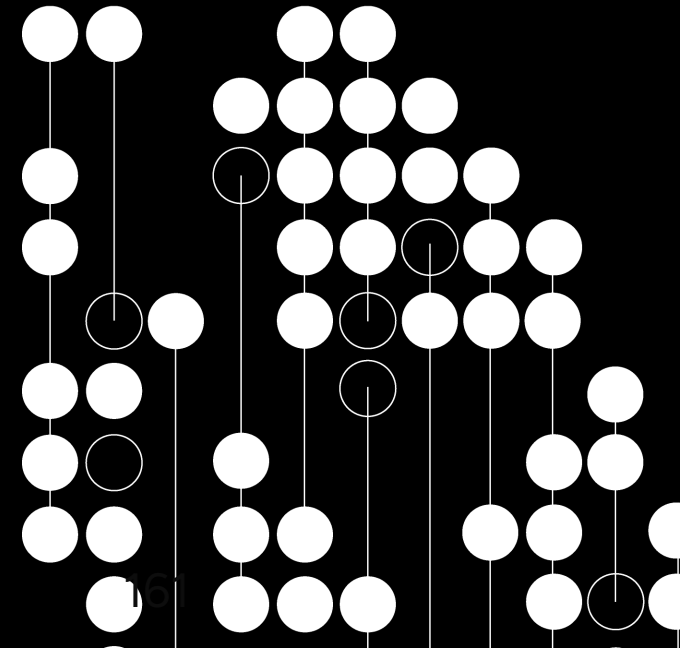
Log files are vital records that document system activities, errors, and events, providing valuable insights into the behaviour of operating systems, applications, and hardware.

Tools like the **Windows event viewer** enable users to access, filter, and analyse these logs for troubleshooting, monitoring, and auditing purposes.

Proper management of log files, including archiving and setting overwrite policies, ensures efficient storage and continuous tracking.

By leveraging log files, IT professionals can identify issues, maintain system security, and optimise performance effectively.

Physical components of a network



Physical components of a network

- The physical components of a network, also known as network hardware, form the foundation for establishing connectivity, enabling data transmission, and ensuring efficient communication between devices.
- These components can be broadly classified into:
 - end devices.
 - connecting devices.
 - transmission media.
 - auxiliary components.

Physical components of a network: End devices

These are the devices at the edges of a network that generate, receive, and process data. They include:

- **Computers:** Desktops, laptops, and servers that participate in data exchange and perform various tasks within the network.
- **Mobile devices:** Smartphones, tablets, and other portable devices that access network resources.
- **Printers and scanners:** Peripheral devices connected to the network to share printing and scanning capabilities.
- **IoT devices:** Smart appliances, sensors, and wearable devices designed for specific applications like home automation, monitoring, or industrial processes.

Physical components of a network: Central connecting devices

These devices ensure the proper routing, management, and delivery of data within the network. They include:

- **Switches:** Connect multiple devices within a LAN and use MAC addresses and IP addresses to forward data packets to the correct destination.
- **Routers:** Operate at Layer 3 (network layer) of the OSI model. Connect different networks (e.g., LAN to WAN) and use IP addresses to determine the optimal path for data transmission.
- **Hubs:** Basic devices that connect multiple computers within a LAN by broadcasting data to all connected devices. Less efficient compared to switches as they do not filter traffic.

Physical components of a network: Other connected devices

- **Modems:** Convert digital data from a computer to analogue signals for transmission over telephone lines and vice versa. Commonly used to provide Internet access in homes and offices.
- **Access Points (APs):** Facilitate wireless connectivity by connecting wireless devices to a wired network. Often integrated into Wi-Fi routers.
- **Firewalls:** Hardware-based firewalls are used to monitor and control incoming and outgoing traffic based on security rules, protecting the network from unauthorised access.

Physical components of a network: Wired transmission media

These are the physical pathways through which data travels within a network. They can be categorised as wired or wireless. These include:

Twisted-Pair Cables:

- Common in LANs, consisting of pairs of insulated copper wires twisted together.
- Types: Unshielded Twisted Pair (UTP) and Shielded Twisted Pair (STP).
- Example: Ethernet cables (e.g., Cat 5, Cat 6).

Coaxial Cables:

- Consist of a central conductor, insulating layer, and metallic shield.
- Used for cable TV networks and older Ethernet networks.

Fiber Optic Cables:

- Use light signals to transmit data, offering high speed and low latency.
- Ideal for long-distance communication and high-bandwidth applications.

Physical components of a network: Wireless transmission media

Wireless transmission media can be broadly categorised based on the type of electromagnetic waves used for communication. These include:

Radio waves:

- Used in Wi-Fi and cellular networks.
- Provide flexible connectivity over a specific range.

Microwaves:

- Utilised for point-to-point communication, such as satellite links.

Infrared:

- Commonly used in remote controls and short-range communication.

Satellite links:

- Enable global communication by transmitting signals to and from satellites.

Physical components of a network: Auxiliary components

These components support the operation, monitoring, and management of the network. These include:

Network Interface Cards (NICs):

- Hardware installed in end devices to enable them to connect to a network.
- Available in wired (Ethernet) and wireless (Wi-Fi) versions.

Repeaters:

- Amplify and regenerate signals to extend the range of a network, ensuring data integrity over long distances.

Bridges:

- Connect two or more LAN segments, allowing communication between them while reducing traffic.

Physical components of a network: Other auxiliary components

Patch panels:

- Centralised hardware that organises and connects network cables in structured cabling systems.

Power over Ethernet (PoE) devices:

- Provide power and network connectivity to devices like IP cameras and access points through Ethernet cables.

Network racks:

- Physical enclosures that house and organise network equipment such as switches, routers, and servers.

Physical components of a network: Power and backup systems

Power and backup systems

To ensure uninterrupted network operation, especially in critical environments, power management is vital. These include:

Uninterruptible Power Supplies (UPS):

- Provide backup power to network devices during outages.

Power Distribution Units (PDUs):

- Distribute power to multiple network devices in a rack.

Physical components of a network

Monitoring and management tools

Devices used for network diagnostics, monitoring, and maintenance. These include:

Network analysers:

- Used to troubleshoot network issues by analysing data packets.

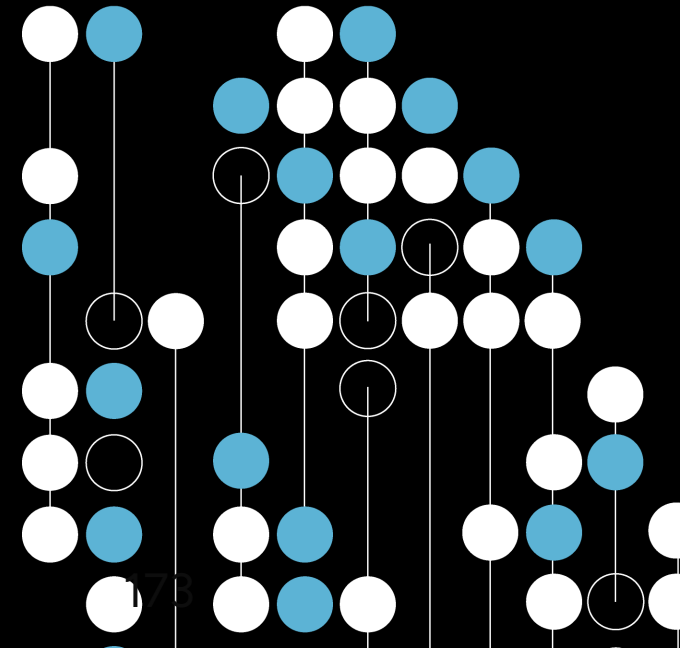
KVM switches:

- Allow control of multiple servers or computers using a single keyboard, video monitor, and mouse.

Physical components of a network: Summary

- Physical components of a network are essential for building and maintaining a reliable, secure, and efficient communication infrastructure.
- Physical components work together to enable data transmission, facilitate connectivity, and ensure smooth network operation.
- Understanding these components helps in designing networks that meet specific performance and scalability requirements.

Types of networks



Types of networks

Networks can range from small-scale setups like **Personal Area Networks (PANs)**, which connect individual devices within a limited range, to large-scale **Wide Area Networks (WANs)** that span across regions and even continents, such as the Internet.

Networks are categorised based on various factors, including:

- size.
- geographical scope.
- structural topology.
- intended purpose.
- underlying technology used for connectivity.

This classification helps in understanding the design, functionality, and application of each type of network in different scenarios.

Types of networks: Personal Area Network (PAN)

- Designed to facilitate communication between devices, such as:
 - smartphones.
 - personal digital assistants.
 - other computing devices, within proximity to an individual.
- May or may not be owned by the individual.
- They enable communication by two methods, either:
 - between the personal devices themselves (intrapersonal communication); or
 - by connecting to a broader network or the Internet via an uplink.

Types of networks: Personal Area Network (PAN)

- A PAN operates within a range of a few meters around the user's workspace, allowing:
 - synchronisation of devices.
 - access to local peripherals like printers.
 - interaction with portable gadgets.
- Technologies commonly used to establish these networks include:
 - IrDA
 - bluetooth
 - USB

Types of networks: Local Area Network (LAN)

- Designed to interconnect computing resources within a geographically limited area.
- Typically confined to a radius of around 10 kilometres or less.
- Commonly used within single rooms, multiple rooms in a building, or across several buildings within a campus or similar premises.
- Enable seamless communication, resource sharing, and data transfer among connected devices such as computers, printers, and servers.
- Advantageous in environments where centralised management and high-speed connectivity are essential, such as:
 - offices.
 - schools.
 - factories.

Types of networks: Metropolitan Area Network (MAN)

- Interconnects computing resources across a metropolitan area, typically covering a city or an urban region.
- Bridges the gap between Local Area Networks (LANs) and Wide Area Networks (WANs).
- Offering connectivity over a larger area than a LAN but smaller than a WAN.
- Can be owned and operated by a single organisation, such as a corporation with distributed offices, but it is more commonly shared among multiple users, including:
 - individuals.
 - businesses.
 - organisations.

Types of networks: Metropolitan Area Network (MAN)

- In some cases, MANs are managed as public utilities, providing essential networking infrastructure to urban communities.
- Examples of MAN applications include:
 - Companies with offices or facilities dispersed throughout a city or county, using the MAN to unify their networks.
 - Public initiatives such as offering free citywide Internet access, connecting residential areas, business districts, and public spaces.

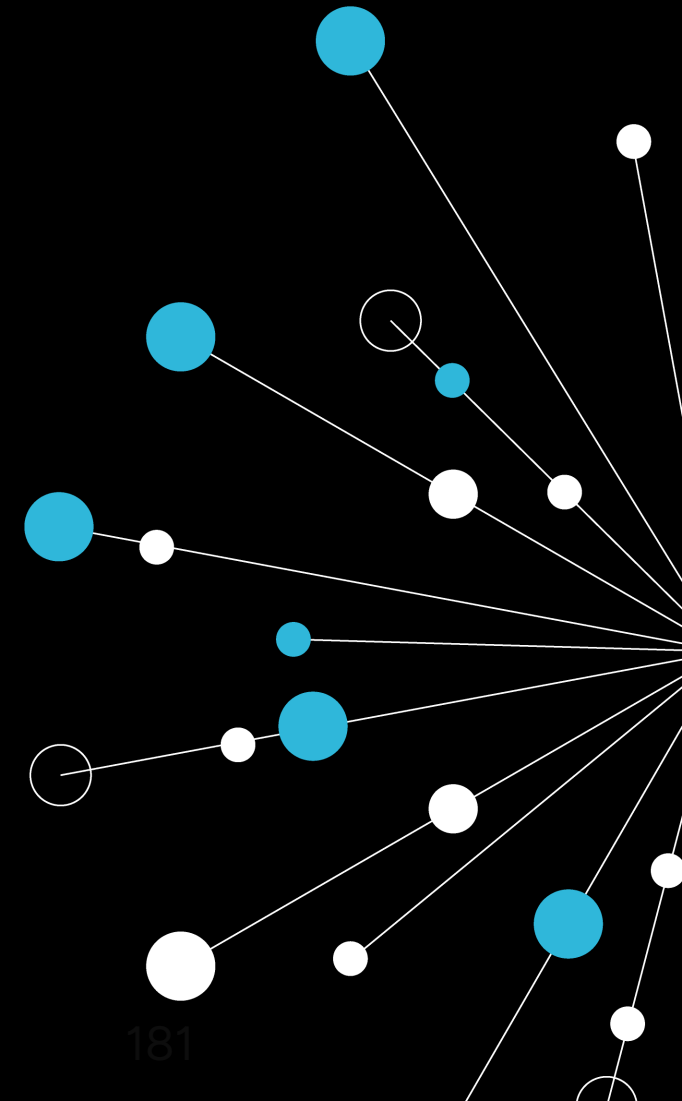
Types of networks: Wide Areas Network (WAN)

- Designed to interconnect computing resources that are spread across vast geographical areas.
- Spans regions such as towns, cities, states, and countries.
- Essential for large-scale communication and data sharing.
- Conceptualised as a collection of connected LANs, e.g., the Internet.
- Require robust infrastructure and advanced technology to handle long-distance communication.
- Most WANs are composed of two primary components that work together to ensure data is transmitted efficiently across the network.

Activity 22: Q&A on types of networks

Complete the activity in your Exercise Guide. Once finished, use Miro to share your answers and findings.

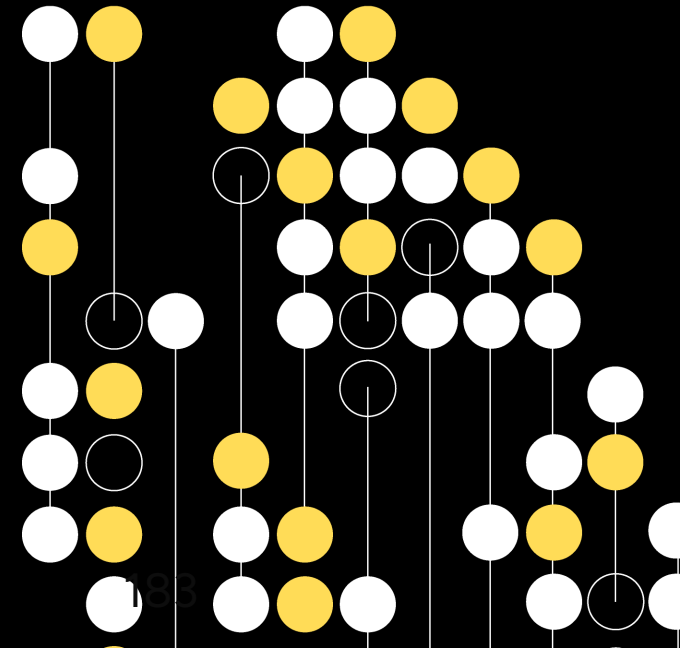
If you need help, feel free to reach out to your tutor.



Physical components of a network: Summary

- The types of area networks, PANs to WANs, highlight the versatility of network technologies in addressing different connectivity needs.
- PANs and LANs cater to localised, small-scale environments.
- MANs, WANs, and Global Area Networks (GANs) extend connectivity to city-wide, regional, and global levels, respectively.
- Emerging technologies like Low-Power Wide-Area Networks (LPWANs) illustrate how networks are evolving to meet the demands of scalability, flexibility, and innovation.
- Network types form the backbone of modern communication, enabling seamless connectivity and collaboration across personal, organisational, and global scales.

Networking models



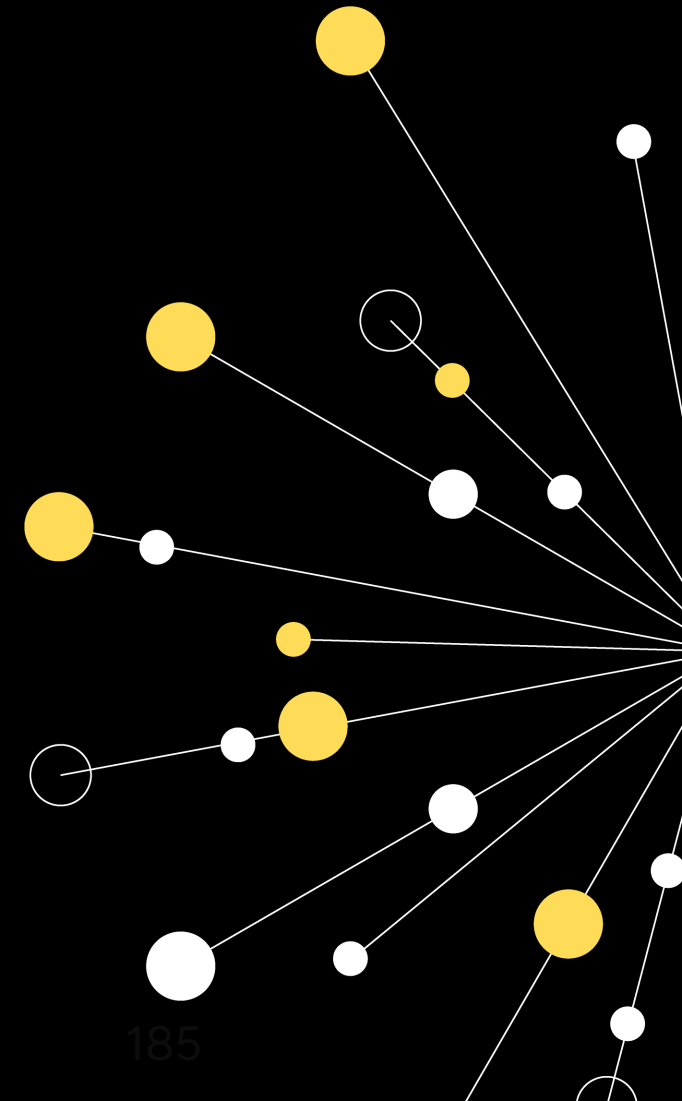
Networking models

- Network models define the structure and framework of communication within a network by specifying how devices, systems, and resources interact.
- They serve as the foundation for organising, implementing, and managing networks based on specific requirements and goals.
- Each model caters to different use cases, ranging from small-scale setups like home networks to large-scale systems like the Internet.

Activity 23: Q&A on networking models

Complete the activity in your Exercise Guide. Once finished, use Miro to share your answers and findings.

If you need help, feel free to reach out to your tutor.



Networking models: The client-server model

- Centralised networking architecture where a central server provides resources, services, or applications to client devices.
- The server acts as the hub that processes requests and distributes information, while the clients rely on the server to access data or resources.

Key features:

- Centralised control and management.
- Servers provide specific services, such as file hosting, email, or database management.
- Clients send requests to the server, which responds with the requested data or action.

Applications:

- Web applications: Browsers (clients) connect to web servers.
- Corporate networks: Centralised file storage, databases, and email services.

Networking models: The client-server model

Advantages:

- Centralised administration simplifies network management.
- High scalability: Additional servers can be added to handle more clients.
- Enhanced security: Servers can enforce consistent policies and monitor activity.

Disadvantages:

- Higher setup and maintenance costs.
- Server failure can disrupt the entire network.

Networking models: Centralised networking model

- All resources, processing, and decision-making are concentrated in a single location, typically a central server or mainframe.
- Devices within the network communicate through the central node.

Key features:

- One central server or system manages all network operations.
- Devices (clients) rely on the central node for resources and instructions.
- Suitable for environments requiring strict control and monitoring.

Applications:

- Legacy mainframe systems.
- Single-location corporate networks.
- Banking systems with centralised data processing.

Networking models: Centralised networking model

Advantages:

- Simplified control and maintenance.
- Easier to implement consistent security policies.
- High performance for local networks.

Disadvantages:

- Single point of failure.
- Not suitable for distributed or global systems.

Networking models: The distributed model

- Spreads resources and tasks across multiple nodes or systems, each capable of operating independently.
- No central authority, and the workload is divided among all participants.

Key features:

- Resources and processing power are spread across the network.
- Nodes can collaborate or function independently.
- Often involves redundancy for fault tolerance.

Applications:

- Content Delivery Networks (CDNs).
- Distributed databases like Cassandra and Hadoop.
- Large-scale applications like Google Search and cloud computing.

Networking models: The distributed model

Advantages:

- Highly scalable and fault-tolerant.
- Efficient resource utilisation by distributing workload.
- No single point of failure.

Disadvantages:

- Complex to design and maintain.
- Requires robust synchronisation mechanisms.

Networking models: The Peer-to-Peer model

- Decentralised architecture where all devices (nodes) in the network have equal status.
- Each device can act as both a client and a server, sharing resources directly with other nodes without requiring a central server.

Key features:

- Direct communication between devices.
- Resources such as files or computing power are shared without a central authority.
- No dependency on a single point of failure.

Applications:

- File-sharing networks: BitTorrent.
- Blockchain technologies and cryptocurrency.
- Small office or home networks.

Networking models: The Peer-to-Peer model

Advantages:

- Cost-effective and easy to set up for small networks.
- Eliminates the need for dedicated servers.
- Scalable for specific use cases, such as file-sharing.

Disadvantages:

- Scalability is limited.
- Lacks centralised security controls.

Networking models: The Hybrid model

- Combines elements of the Client-Server and Peer-to-Peer architectures.
- While central servers provide essential services, nodes can also communicate directly for specific tasks.

Key features:

- Centralised control for critical services.
- Peer-to-peer interactions for direct communication.
- Flexibility to adapt to various scenarios.

Applications:

- Messaging platforms like Microsoft Teams.
- Cloud-based collaboration tools with local caching.
- Gaming networks where servers manage matchmaking, but players connect directly.

Networking models: The Hybrid model

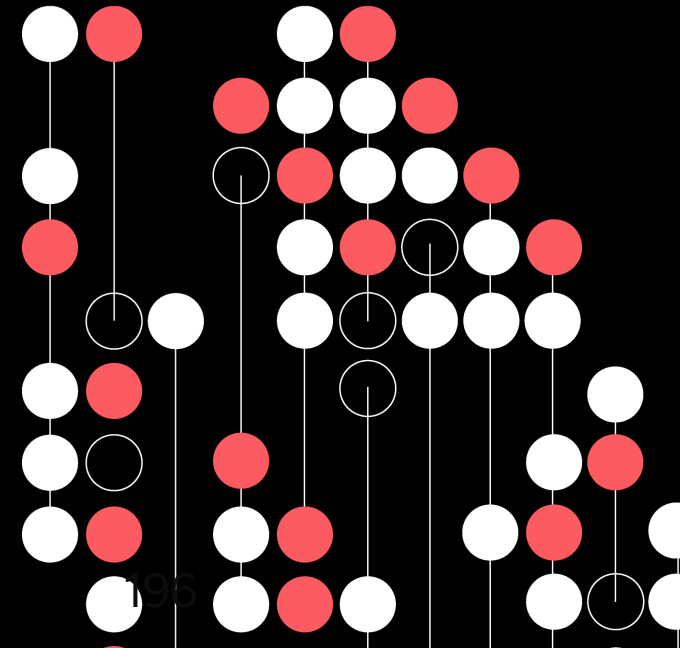
Advantages:

- Combines the strengths of centralised and decentralised models.
- Efficient for large-scale networks with diverse requirements.
- Balances resource usage and performance.

Disadvantages:

- Can be complex to manage.

Wireless networking



Wireless networking

Recap from Module 3 Core Cloud & Networking Skills:

- History of Wi-Fi
- 802.11 Standards
- 2.4Ghz vs 5Ghz
- 2.4Ghz Channels
- Wi-Fi Security Protocols
- Wi-Fi Enterprise Mode

Wireless networking

How Wi-Fi Works

Relies on advanced data encoding and modulation techniques to ensure that information is transmitted efficiently and reliably over wireless channels.

- **Encoding:** Involves converting digital data into a format suitable for transmission as radio signals.
- **Modulation:** Refers to altering these signals in specific ways to carry the encoded data effectively.

Wireless networking: Orthogonal Frequency Division Multiplexing (OFDM)

OFDM in Wi-Fi: OFDM is a key modulation technique used in Wi-Fi standards, starting with 802.11a and continuing with Wi-Fi 6 (802.11ax).

How OFDM works:

- Splits high-speed data into smaller sub-streams.
- Transmits these sub-streams over several subcarriers within a frequency channel.

Wi-Fi frequency bands:

- Wi-Fi operates on 2.4 GHz, 5 GHz, and 6 GHz bands.
- Each band is divided into multiple channels.

OFDM channel components:

- **Data subcarriers:** Carry user data packets.
- **Pilot subcarriers:** Used for synchronisation and error correction.
- **Guard subcarriers:** Prevent interference between adjacent channels.

Wireless networking: Wi-Fi 6

Released in 2019, designed to improve performance in dense environments like offices, stadiums, and smart homes.

- **Higher data speeds:** Maximum theoretical speed of up to 9.6 Gbps (compared to 3.5 Gbps in Wi-Fi 5) using advanced modulation (1024-QAM) and channel bonding.
- **Increased spectrum usage:** Operates on 2.4 GHz, 5 GHz, and 6 GHz bands (Wi-Fi 6E), offering more channels and reducing congestion.
- **Better security:** Supports WPA3, offering stronger encryption and protection against brute-force attacks.
- **Advantages of Wi-Fi 6:** Ideal for high-density environments like smart homes, offices, and public spaces. Better streaming and gaming experiences with lower latency.

Wireless networking: Wi-Fi 7

The next-generation standard, expected to become widely available starting in 2024.

- **Groundbreaking features:**
 - Delivering extreme speeds and ultra-low latency.
 - Meeting the demands of emerging technologies like AR/VR, 8K streaming, and cloud gaming.
- **Unprecedented speed:** Maximum theoretical speed of up to 46 Gbps, nearly 5x faster than Wi-Fi 6. Enabled by wider channels, higher QAM, and improved technologies.
- **Extremely wide channels:** Uses 320 MHz channels, double the width of Wi-Fi 6 channels (160 MHz) – providing more bandwidth for faster data transmission.

Wireless networking: Wi-Fi mesh networks

- Modern solution designed to eliminate dead zones and provide seamless, reliable wireless connectivity over large areas.
- Unlike traditional single-router systems, mesh network use multiple devices (called nodes) working together to blanket an area with consistent Wi-Fi coverage.
- Beneficial for large homes, multi-story buildings, offices, and outdoor areas.
- A mesh network is made up of:
 - a primary node (often connected to a modem).
 - secondary nodes (satellites or extenders).
- Nodes communicate with each other wirelessly or via wired connections to create a unified network.
- All nodes broadcast the same SSID (network name), for seamless roaming.

Wireless networking: Radio Frequency Interference (RFI)

- Occurs when unwanted radio signals disrupt wireless network operations, including Wi-Fi.
- Wi-Fi operates using radio waves in the 2.4 GHz and 5 GHz frequency bands, interference from other devices or signals in these ranges can degrade network performance, causing issues such as:
 - reduced speed.
 - dropped connections.
 - inconsistent coverage.
- Devices in the same frequency bands can interfere with Wi-Fi signals, including:
 - Microwaves (2.4 GHz)
 - Cordless phones
 - Bluetooth devices
 - Baby monitors
 - Wireless cameras

Wireless networking: Wi-Fi troubleshooting

Wi-Fi issues can include slow speeds, intermittent connections complete outages. Troubleshooting involves identifying causes and applying solutions.

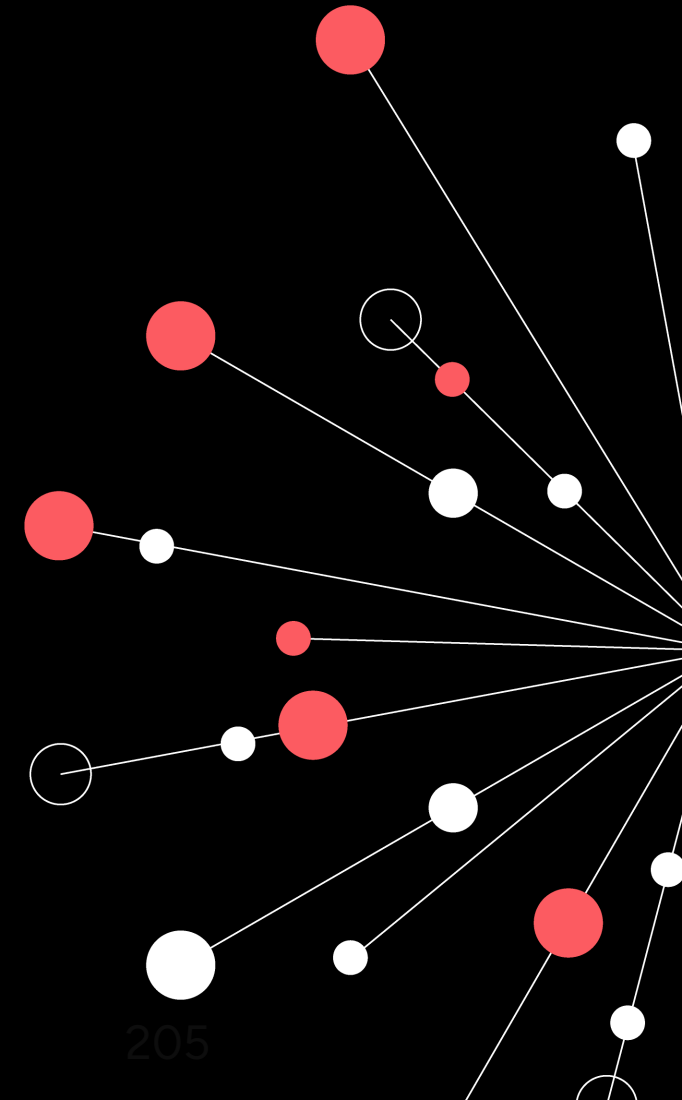
Common Wi-Fi issues and troubleshooting methods:

1. Ensure Wi-Fi is turned on.
2. Disable Airplane mode.
3. Check the Wi-Fi network appears in the list of networks.
4. If connected, disconnect and reconnect.
5. Try connecting on a different frequency band (2.4GHz and 5GHz).
6. Make sure physical Wi-Fi switch on your laptop is turned on.
7. Run the network troubleshooter.
8. Restart your modem and wireless router.

Activity 24: Q&A on wireless networking

Complete the activity in your Exercise Guide. Once finished, use Miro to share your answers and findings.

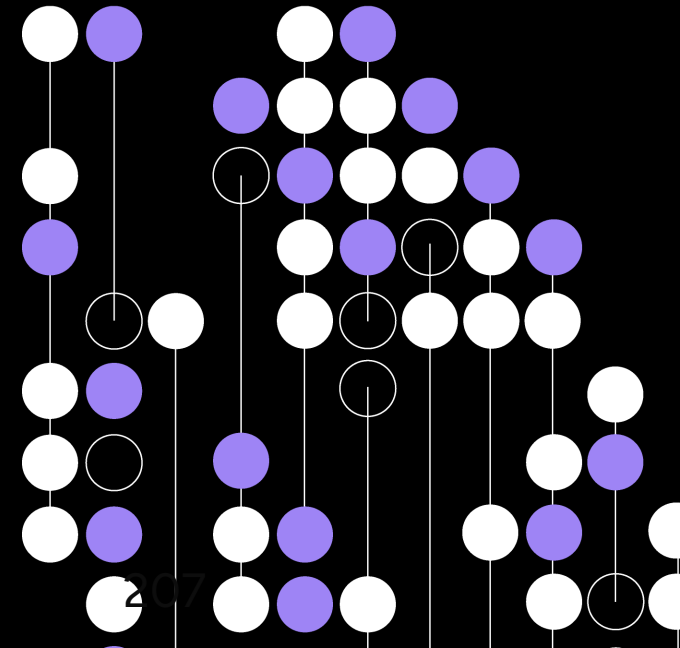
If you need help, feel free to reach out to your tutor.



Wireless networking: Summary

- Wi-Fi is essential for modern connectivity, but its performance can be affected by factors like Radio Frequency Interference (RFI).
- Mitigating RFI through proper router placement and frequency optimisation ensures efficient networks.
- Advances in Wi-Fi 6 and 7 bring faster speeds, lower latency, and support for more devices, making them ideal for high-density environments.
- Effective troubleshooting, such as checking connectivity and reducing interference, helps resolve common issues.
- Regular updates, managing network congestion, and using mesh systems enhance reliability.
- Staying informed about Wi-Fi technologies and applying these strategies ensures robust and seamless wireless connectivity.

Principles of network addressing



Principles of network addressing

- A fundamental concept in computer networks that allows devices to identify and communicate with each other.
- Provides a structured and systematic way to assign unique identifiers to devices within a network, ensuring:
 - efficient data routing.
 - management.
 - communication.

Principles of network addressing: Lesson coverage

The key principles of network addressing:

- Uniqueness
 - Hierarchical structure
 - Scalability – IPv4 vs. IPv6
 - Address types
 - Address allocation
- 
- (Recap)**

IPV4:

- Address resolution
- Subnetting
- Address classes and CIDR
- Dynamic and static assignment

Principles of network addressing: Uniqueness

- Fundamental to the proper functioning of any network.
- Each device within a network must be assigned a distinct address to avoid conflicts and ensure that data is accurately delivered to the intended recipient.

IPv4 address structure:

- Consists of 32 bits, typically represented in dotted decimal format (e.g., 192.168.1.1).
- The 32 bits allow for approximately **4.3 billion unique addresses**, which initially seemed sufficient but has become limited due to the rapid growth of internet-connected devices.
- To manage address scarcity, IPv4 introduced private address ranges (e.g., 192.168.x.x) that can be reused within local networks.
- These are made unique globally through technologies like Network Address Translation (NAT).

Principles of network addressing: Hierarchical structure

IPv4 addresses are 32-bit numerical identifiers, typically represented in dotted decimal notation (e.g., 192.168.1.1).

These addresses are inherently hierarchical, consisting of two primary parts:

1. Network ID (network prefix):

- Identifies the specific network to which the device belongs.
- All devices within the same network share the same network ID.
- Routers use the network ID to determine where to forward packets on their journey to the destination network.

2. Host ID (host portion):

- Identifies the specific device within the network.
- Each device on the same network must have a unique host ID.

Principles of network addressing: Scalability

- A critical principle of network addressing.
- Ensures networks can grow and evolve to accommodate an increasing number of devices, applications, and users.
- A scalable addressing system must handle this growth seamlessly without compromising performance, usability, or efficiency.
- IPv4, while foundational to the modern internet, has encountered significant scalability challenges, leading to the development of IPv6 and other solutions to address its limitations.

Principles of network addressing: Address types

In IPv4 addressing, different address types facilitate diverse communication scenarios within networks, each serving a specific purpose to ensure efficient data delivery and support various networking needs:

- **Unicast:**
 - The most common form of communication in IPv4 networks.
 - Involving one-to-one communication.
 - A single sender transmits data to a single, specific recipient.
- **Multicast:**
 - Enables one-to-many communication.
 - A single sender transmits data to multiple specific recipients simultaneously.
- **Broadcast:**
 - Allows one-to-all communication.
 - A single sender transmits data to all devices within a specific network segment.

Principles of network addressing: Address allocation

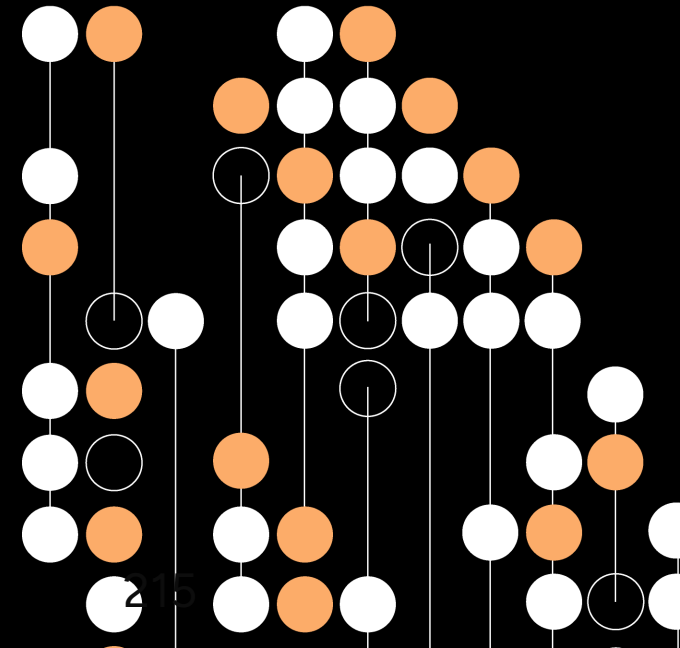
- The process of assigning IPv4 addresses to devices within a network.
- Ensures that each device has a unique identifier to enable seamless communication.
- Efficient address allocation is essential for optimising the limited IPv4 address space, minimising conflicts, and ensuring smooth network operations.

Examples of IPv4 allocation:

- Host and Network Breakdown
- Classes
- APIPA
- Public and Private Allocation
- Static and Dynamic Address Allocation
CIDR

IPv4

QA



IPv4 recap:

- **Host and network breakdown:**

Default subnet masks help defined what octet within the IP are a Network and Host.

- 255.0.0.0
- 255.255.0.0
- 255.255.255.0

- **IP address classes:**

- **Class A:** 0.0.0.0 to 127.255.255.255
- **Class B:** 128.0.0.0 to 191.255.255.255
- **Class C:** 192.0.0.0 to 223.255.255.255
- **Class D:** 224.0.0.0 to 239.255.255.255 (Multicast)
- **Class E:** 240.0.0.0 to 255.255.255.255 (Reserved)

- **APIPA (Automatic Private IP Addressing):**

- If a device can't obtain an address from a DHCP server, it self-assigns one in the 169.254.x.x range.

- **Loopback address:**

- 127.x.x.x is used to verify network stack functionality without requiring external network hardware.

- **Public and private allocation, common private ranges:**

- 10.0.0.0 to 10.255.255.255
- 172.16.0.0 to 172.31.255.255
- 192.168.0.0 to 192.168.255.255

IPv4: Address resolution

- The process of mapping an IP address (logical address) to a corresponding MAC address (physical address) within a local network.
- This ensures that data packets can be transmitted over the physical layer of a network. IPv4 employs the Address Resolution Protocol (ARP) to achieve this mapping.
- **Logical address** (IP address): IPv4 uses a 32-bit logical address to identify devices on a network. Logical addresses are used for routing data between networks.
- **Physical address** (MAC address): A MAC address is a hardware-specific, 48-bit unique identifier assigned to network interfaces. Required for data transmission within a local area network (LAN).
- **Need for address resolution:** While IP addresses are sufficient for logical identification and routing, the actual delivery of packets within a LAN requires the MAC address of the destination device.

IPv4: Address resolution

- When a device wants to communicate with another device on the same LAN, it checks its ARP table (a cache of known IP-to-MAC mappings).
- If the desired IP address is not in the ARP table, the device broadcasts an ARP Request packet to all devices on the local network.
- The ARP Request contains the sender's IP and MAC addresses, as well as the target IP address.
- To view the ARP table on a device, you can use the following command:
`arp -a`
To clear the ARP cache (if needed): Use `arp -d` to delete all entries or
- `arp -d <IP>` for a specific entry.

IPv4: Subnetting

- Subnetting and aggregation are two essential techniques used in network design to manage:
 - efficient IP address allocation.
 - optimised routing.
 - improved scalability.
- **Subnetting** is the process of dividing a larger network into smaller, more manageable segments, called subnets. It involves borrowing bits from the host portion of an IP address to create additional network identifiers.

IPv4: Subnetting

- To break networks down from their original classes, refer to the chart below for the relevant properties:

Class C IP	192	168	1	1
Default subnet mask	255	255	255	0
	N	N	N	H
	11111111	11111111	11111111	00000000

- We can see how the default subnet mask defines the Network and Host portions of the IP address.
- When we introduce subnetting we will be focusing on the **.0** portion of the subnet mask and change its value to make the network smaller.

IPv4: Subnetting

- The table below breaks down the various subnets masks we can create.
- You can use the conversion chart to help you convert each row from binary into decimal.

Subnet masks in decimal				Subnet mask in binary				
255	255	255	0		11111111	11111111	11111111	00000000
255	255	255	128		11111111	11111111	11111111	10000000
255	255	255	192		11111111	11111111	11111111	11000000
255	255	255	224		11111111	11111111	11111111	11100000
255	255	255	240		11111111	11111111	11111111	11110000
255	255	255	248		11111111	11111111	11111111	11111000
255	255	255	252		11111111	11111111	11111111	11111100
255	255	255	254		11111111	11111111	11111111	11111110
255	255	255	255		11111111	11111111	11111111	11111111

IPv4: Subnetting

- By focusing on the **last octet** and incorporating our conversion chat, this is what the layout would look like:

Subnet masks decimal				Subnet mask in binary					128	64	32	16	8	4	2	1
255	255	255	0	11111111	11111111	11111111	00000000		0	0	0	0	0	0	0	0
255	255	255	128	11111111	11111111	11111111	10000000		1	0	0	0	0	0	0	0
255	255	255	192	11111111	11111111	11111111	11000000		1	1	0	0	0	0	0	0
255	255	255	224	11111111	11111111	11111111	11100000		1	1	1	0	0	0	0	0
255	255	255	240	11111111	11111111	11111111	11110000		1	1	1	1	0	0	0	0
255	255	255	248	11111111	11111111	11111111	11111000		1	1	1	1	1	0	0	0
255	255	255	252	11111111	11111111	11111111	11111100		1	1	1	1	1	1	0	0
255	255	255	254	11111111	11111111	11111111	11111110		1	1	1	1	1	1	1	0
255	255	255	255	11111111	11111111	11111111	11111111		1	1	1	1	1	1	1	1

IPv4: Subnetting

- We now add another heading and titles on top of our conversion chat.
- This will now tell us the **number of Subnets** we can create based on the **final 1 bit**.
- Notice that if you were to multiply each **“No of subnets”** collum with the **“No of Addresses”** you will always get **256** e.g.: **2 X 128, 4 X 64, 8 X 32**

									No of subnets	2	4	8	16	32	64	128	256	No of streets
Subnet masks decimal					Subnet mask in binary				No of addresses	128	64	32	16	8	4	2	1	No of plots
255	255	255	0		11111111	11111111	11111111	00000000		0	0	0	0	0	0	0	0	
255	255	255	128		11111111	11111111	11111111	10000000		1	0	0	0	0	0	0	0	
255	255	255	192		11111111	11111111	11111111	11000000		1	1	0	0	0	0	0	0	
255	255	255	224		11111111	11111111	11111111	11100000		1	1	1	0	0	0	0	0	
255	255	255	240		11111111	11111111	11111111	11110000		1	1	1	1	0	0	0	0	
255	255	255	248		11111111	11111111	11111111	11111000		1	1	1	1	1	0	0	0	
255	255	255	252		11111111	11111111	11111111	11111100		1	1	1	1	1	1	0	0	
255	255	255	254		11111111	11111111	11111111	11111110		1	1	1	1	1	1	1	0	
255	255	255	255		11111111	11111111	11111111	11111111		1	1	1	1	1	1	1	1	

IPv4: Subnetting

- **What does it all mean?** If you had 100 devices in your network:
- The default Class C subnet mask **255.255.255.0** supports 254 usable devices (256 total, minus 2 for Network ID and Broadcast).
- But, by using **255.255.255.128**, you can create a smaller subnet with 126 usable addresses (128 total, minus 2 for Network ID and Broadcast), efficiently managing your address space.

								No of Subnets	2	4	8	16	32	64	128	256	No of Streets				
Subnet masks Decimal				Subnet Mask in Binary				No of Addresses	128	64	32	16	8	4	2	1	No of Plots		No of Addres ses	No of Hosts	No of Netwo rks
255	255	255	0	11111111	11111111	11111111	00000000		0	0	0	0	0	0	0	0			256	254	1
255	255	255	128	11111111	11111111	11111111	10000000		1	0	0	0	0	0	0	0			128	126	2
255	255	255	192	11111111	11111111	11111111	11000000		1	1	0	0	0	0	0	0			64	62	4
255	255	255	224	11111111	11111111	11111111	11100000		1	1	1	0	0	0	0	0			32	30	8
255	255	255	240	11111111	11111111	11111111	11110000		1	1	1	1	0	0	0	0			16	14	16
255	255	255	248	11111111	11111111	11111111	11111000		1	1	1	1	1	0	0	0			8	6	32
255	255	255	252	11111111	11111111	11111111	11111100		1	1	1	1	1	1	0	0			4	2	64
255	255	255	254	11111111	11111111	11111111	11111110		1	1	1	1	1	1	1	0			2	0	128
255	255	255	255	11111111	11111111	11111111	11111111		1	1	1	1	1	1	1	1			1	0	256

IPv4: Subnetting

What does it mean?

If we use the IP address 192.168.1.0 along with the subnet mask 255.255.255.128 our break down will be as follows:

	Subnet A	Subnet B	
Network ID	192.168.1.0	192.168.1.128	Street Sign
First Host	192.168.1.1	192.168.1.129	First House
Last Host	192.168.1.126	192.168.1.254	Last House
Broadcast	192.168.1.127	192.168.1.255	Noticeboard

IPv4: Subnetting

- In another example, if we use the same IP address 192.168.1.0 along with the subnet mask 255.255.255.192, our break down will be as follows:

	Subnet A	Subnet B	Subnet C	Subnet D	
Network ID	192.168.0.0	192.168.0.64	192.168.0.128	192.168.0.192	Street Sign
First Host	192.168.0.1	192.168.0.65	192.168.0.129	192.168.0.193	First House
Last Host	192.168.0.62	192.168.0.126	192.168.0.190	192.168.0.254	Last House
Broadcast	192.168.0.63	192.168.0.127	192.168.0.191	192.168.0.255	Noticeboard

- If we refer to the chart on slide 224 you will notice that we have:
4 Subnets with 64 Addresses of which we will have 62 Hosts.

IPv4: Subnetting summary

Subnetting is a versatile and practical tool that optimises IP address usage and enhances network management in various environments, including:

- households.
- workplaces.
- larger network environments.

Households: Subnetting can be employed to create separate subnets for different types of devices or users:

- Allocate **one subnet for personal devices** (smart TVs, laptops, gaming consoles, etc.) to ensure seamless connectivity and communication.
- Allocate **another subnet for guest users**, providing internet access while isolating them from personal devices and sensitive data – enhancing security and better control over network traffic.
- Guests enjoy internet access without interfering with or compromising your private network.

IPv4: Subnetting summary

Workplaces: Similar principles (as households) can be applied to segregate departments or functional areas, maintaining data privacy and optimising performance for specific groups.

Subnetting is an essential technique for modern network management, by implementing it, both home and professional environments can benefit from:

- improved organisation.
- enhanced security.
- tailored connectivity.

IPv4: Address Classes and CIDR

- **CIDR**, or **Classless Inter-Domain Routing**, is a method for allocating IP addresses and routing that replaces the traditional class-based IP addressing system.
- Introduced in 1993, CIDR was developed to address inefficiencies in IP address allocation and to combat the rapid exhaustion of IPv4 addresses.
- In CIDR, an IP address is followed by a slash (/) and a number, which represents the number of bits used for the network prefix.
 - **Example:** 192.168.0.0/**24** indicates a network with a **24-bit** network prefix (255.255.255.0).
 - In simple terms, the slash indicator is the number of **1 bits** used in the subnet mask.

IPv4: Address Classes and CIDR

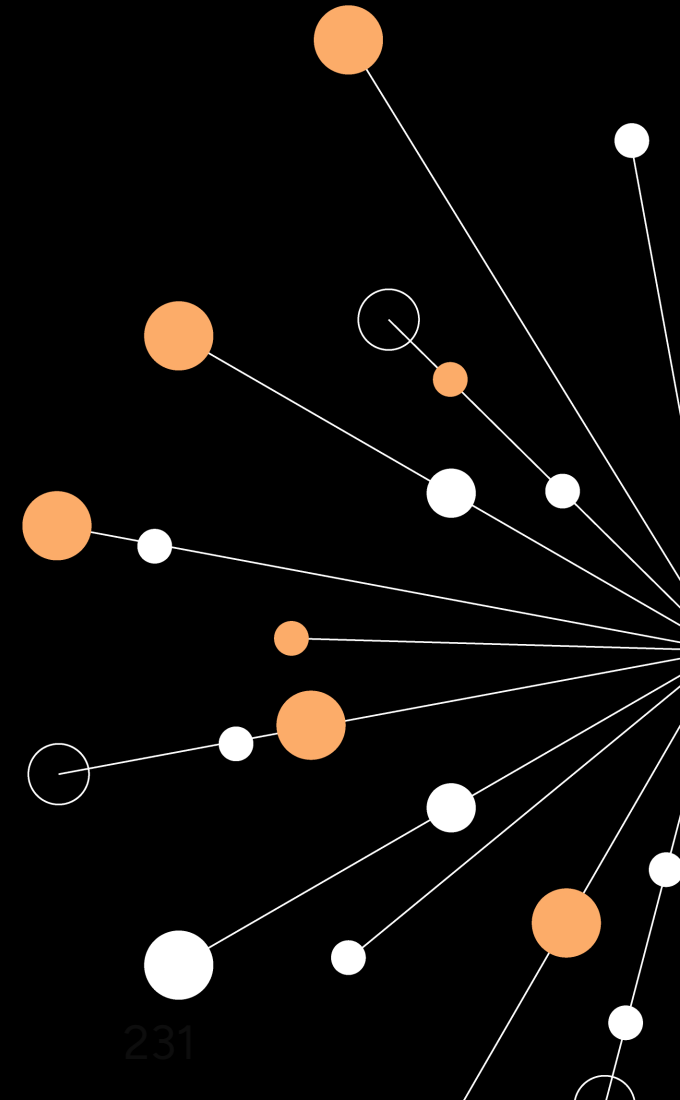
- Using CIDR, an IP like 192.168.1.0/28 indicates that 28 bits are used for the subnet mask, corresponding to 255.255.255.240.
- This configuration provides 16 addresses per subnet, allowing for 14 usable hosts across 16 subnets.

									No of Subnets	2	4	8	16	32	64	128	256	No of Streets			
CIDR /	Subnet Mask in Binary				Subnet masks Decimal				No of Addresses	128	64	32	16	8	4	2	1	No of Plots	No of Addresses	No of Hosts	No of Networks
24	11111111	11111111	11111111	00000000	255	255	255	0		0	0	0	0	0	0	0	0		256	254	1
25	11111111	11111111	11111111	10000000	255	255	255	128		1	0	0	0	0	0	0	0		128	126	2
26	11111111	11111111	11111111	11000000	255	255	255	192		1	1	0	0	0	0	0	0		64	62	4
27	11111111	11111111	11111111	11100000	255	255	255	224		1	1	1	0	0	0	0	0		32	30	8
28	11111111	11111111	11111111	11110000	255	255	255	240		1	1	1	1	0	0	0	0		16	14	16
29	11111111	11111111	11111111	11111000	255	255	255	248		1	1	1	1	1	0	0	0		8	6	32
30	11111111	11111111	11111111	11111100	255	255	255	252		1	1	1	1	1	1	0	0		4	2	64
31	11111111	11111111	11111111	11111110	255	255	255	254		1	1	1	1	1	1	1	0		2	0	128
32	11111111	11111111	11111111	11111111	255	255	255	255		1	1	1	1	1	1	1	1		1	0	256

Activity 25: Practical on subnetting

Complete the activity in your Exercise Guide. Once finished, use Miro to share your answers and findings.

If you need help, feel free to reach out to your tutor.



IPv4: Dynamic and static assignment

In IPv4 networking, IP address assignment determines how devices receive their unique addresses to communicate within a network.

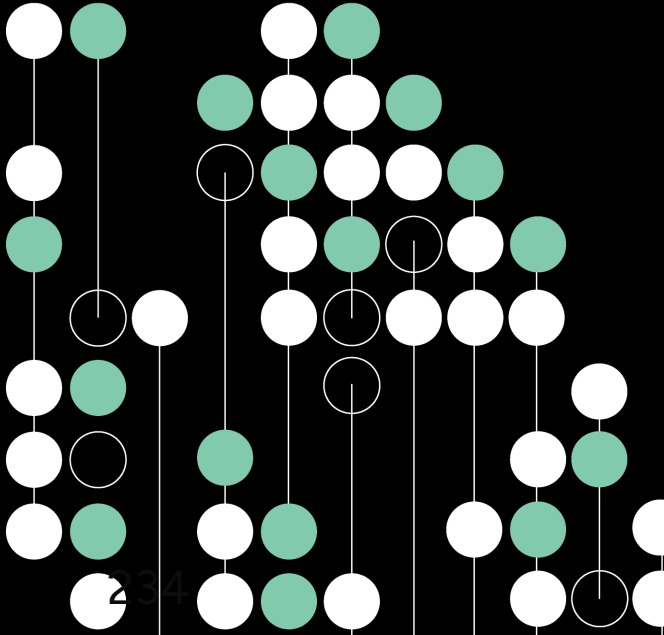
Two primary methods of assignment are **dynamic** and **static**, each serving different purposes and offering distinct advantages.

1. **Dynamic IP assignment:** Automatically allocates an IP address to a device using a protocol like the Dynamic Host Configuration Protocol (DHCP), commonly used in home and enterprise networks for ease of management.
2. **Static IP assignment:** Manually assigns a fixed IP address to a device. This address does not change unless explicitly reconfigured by an administrator.

IPv4: Summary

- IPv4 address allocation has been a foundational aspect of the internet's growth and functionality.
- Initially, IPv4 addresses were plentiful, but as the internet expanded, concerns about address depletion began to emerge.
- The introduction of techniques have helped to extend the usable lifespan of IPv4 addresses, such as:
 - subnetting.
 - Classless Inter-Domain Routing (CIDR).
 - Network Address Translation (NAT).
- With the increasing number of connected devices and the exhaustion of available IPv4 addresses, the need for IPv6 has become apparent.

IPv6



IPv6

Lesson overview:

Recap from Module 3

- Understanding the use of Hexadecimal in IPv6
- IPv6 Address Types and their functions
- IPv6 Address Breakdown

Topics:

- Compression
- Expansion

IPv6: Compression

IPv6 compression involves shortening an IPv6 address by removing redundant elements. The rules for compression are:

1. Remove leading zeros in each group:

- A group (16-bit segment) in an IPv6 address can drop leading zeros.
- Example: **005f** becomes **5f**.

2. Use double colons (::) to replace consecutive groups of zeros:

- If there are one or more consecutive groups of **0000**, replace them with **::** and it can only appear once in an address to avoid ambiguity.
- **Example:**
2001:0db8:0000:0000:0000:ff00:0042:8329 becomes
2001:db8::ff00:42:8329

IPv6: Compression

- The table shows the breakdown of an address from its full form to its compressed form.
- All three examples are valid representations.
- You can ping any of them, and they will respond as expected.

Description	Example
Full IPv6 address	fe80:0000:0000:0000:0184:e9f8:a71b:0304
IPv6 address using zero compression	fe80::0184:e9f8:a71b:0304
IPv6 address using zero compression and dropping leading zeros	fe80::184:e9f8:a71b:304

IPv6: Expansion

IPv6 expansion involves fully writing out a compressed IPv6 address by restoring omitted zeros and replacing :: with the appropriate number of 0000 groups. The rules for expansion are:

1. Expand groups with missing zeros:

- Add leading zeros to each group so that every group is 4 hexadecimal digits long.
- **Example:** 5f becomes 005f. Replace :: with enough 0000 groups:

2. Count how many groups are missing and insert 0000 as placeholders:

- **Example of expansion:**
Compressed: 2001:db8::ff00:42:8329 becomes
2001:db8:0000:0000:0000:ff00:0042:8329

IPv6: Expansion

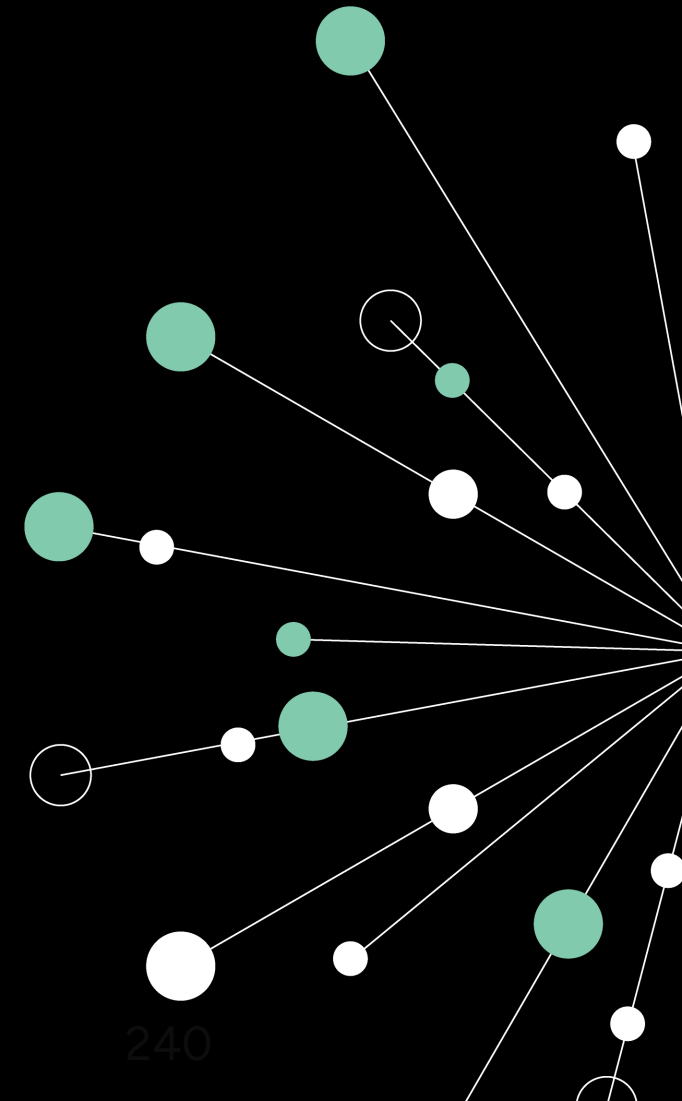
- The table shows the breakdown from the compressed IPv6 address to its expanded form.
- An IPv6 address consists of **8 hextets**, with **each hextet containing 4 hexadecimal digits**.
- This breakdown helps visualise how a compressed address expands to its full, standard form.

Compressed Address	Expanded Address
2001:db8::	2001:0db8:0000:0000:0000:0000:0000:0000
::1	0000:0000:0000:0000:0000:0000:0000:0001
2001:db8:0:0::1	2001:0db8:0000:0000:0000:0000:0000:0001
2001:db8::ff:0:42	2001:0db8:0000:0000:0000:00ff:0000:0042

Activity 26: Q&A on IPv6

Complete the activity in your Exercise Guide. Once finished, use Miro to share your answers and findings.

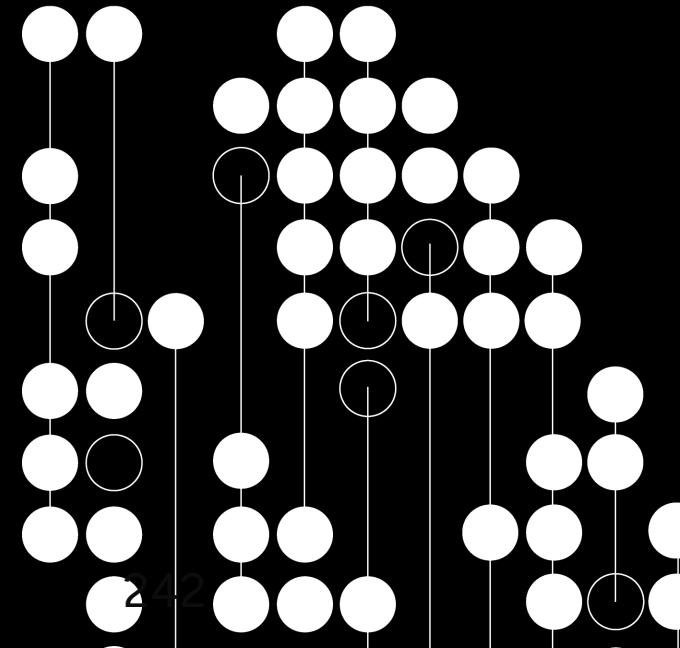
If you need help, feel free to reach out to your tutor.



Compression and expansion: Summary

- Compression and expansion **do not change** the address itself; they only affect its representation.
- When compressing, always aim for the shortest possible representation.
- When expanding, always restore the full 8 groups with exactly 4 hexadecimal digits in each.
- By applying these rules, IPv6 addresses can be efficiently displayed and correctly interpreted.

Computer Names and Services



Computer names and network services

Computer names and services are critical components in computer networking, enabling systems and applications to identify and communicate with each other efficiently.

Here's an expanded look into both:

1. **Computer names:**

- Also known as hostnames, are human-readable labels assigned to devices in a network.
- Names simplify identification and communication, eliminating the need to remember complex IP addresses.

2. **Services:**

- Refer to the processes, applications, or functions running on computers that provide specific capabilities to other devices or users in a network.
- They are accessed using service names or port numbers.

Aspects of computer names

Hostnames:

- A hostname is typically a single-word or a dot-separated string.
- **Example:** **mycomputer** or **server01.local**.
- In larger networks or the internet, hostnames often include a domain (Fully Qualified Domain Name, or FQDN), such as www.example.com.

Usage:

- Used to identify a specific computer within a network.
- Simplifies tasks like file sharing, printer access, or connecting to servers.

DNS integration:

- Hostnames are resolved to IP addresses using the Domain Name System (DNS).

Example: Typing example.com in a browser triggers a DNS query that resolves the name to its corresponding IP address, such as 93.184.216.34.

NetBIOS names:

- In legacy Windows environments, NetBIOS names are used for identifying devices on local networks.

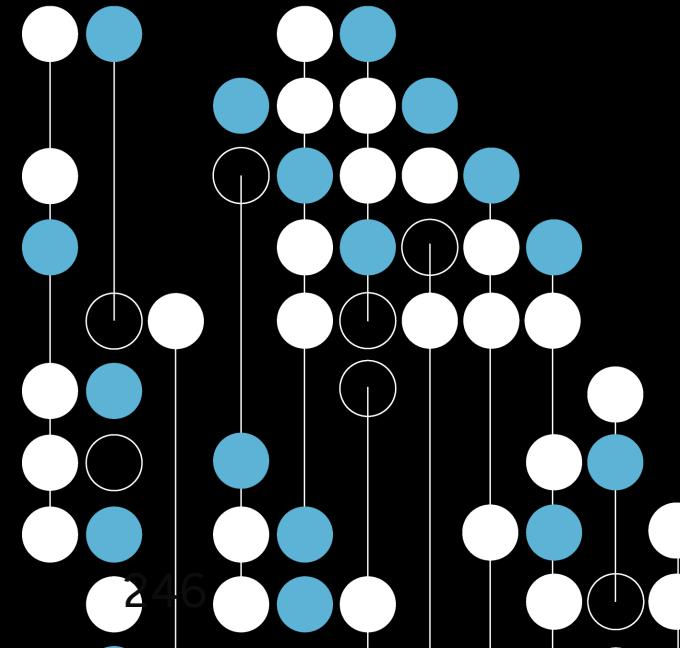
Hostnames and service names

- **Hostnames and service names** make networks more user-friendly and easier to navigate.
- They facilitate seamless interactions between devices, applications, and users while simplifying troubleshooting by offering clear, descriptive identifiers instead of raw IP addresses or port numbers.

Real-world applications:

- **Accessing shared resources:** Instead of typing an IP address, users access shared drives or printers using names like **fileserver** or **printer01.Web**
- **Browsing:** Hostnames such as **www.google.com** allow users to access websites without knowing their IP addresses.
- **Service monitoring:** Network administrators monitor critical services (e.g., web servers, email servers) using service names and associated ports.

Network Services



Network services

- Specialised applications or functions provided over a network to:
 - facilitate communication.
 - share resources.
 - execute specific tasks.
- Services run on servers or devices and are accessed by clients or users within the network.
- Essential for ensuring the smooth operation of modern networks by enabling various activities, such as:
 - browsing the web.
 - sharing files.
 - sending emails.

Network services: Characteristics of network services

- **Protocol-based:**
 - Network services operate using specific communication protocols, such as HTTP, SMTP, FTP, or DNS.
 - These protocols define how data is transmitted and received.
- **Client-server model:**
 - Most network services follow a client-server architecture.
 - The server hosts the service, and the client accesses it by sending requests.
- **Port numbers:**
 - Services are identified by unique port numbers associated with the server's IP address.
 - Example: HTTP (port 80), HTTPS (port 443), SMTP (port 25).

Network services

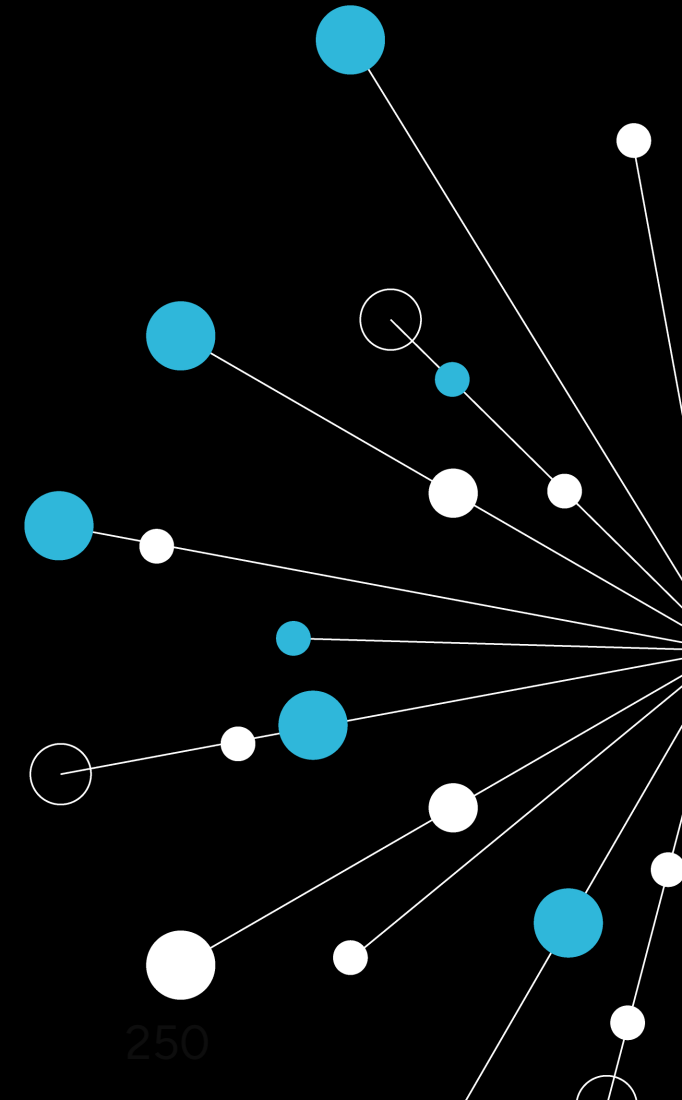
Network services form the backbone of modern connectivity, enabling the efficient and reliable operation of networks for personal, enterprise, and global applications.

- **Enterprise networks:** Use directory and file-sharing services to streamline operations and collaboration.
- **Home networks:** Rely on services like DHCP and DNS for connectivity and web access.
- **Cloud computing:** Provides scalable infrastructure services for businesses and developers.

Activity 27: Q&A on network services

Complete the activity in your Exercise Guide. Once finished, use Miro to share your answers and findings.

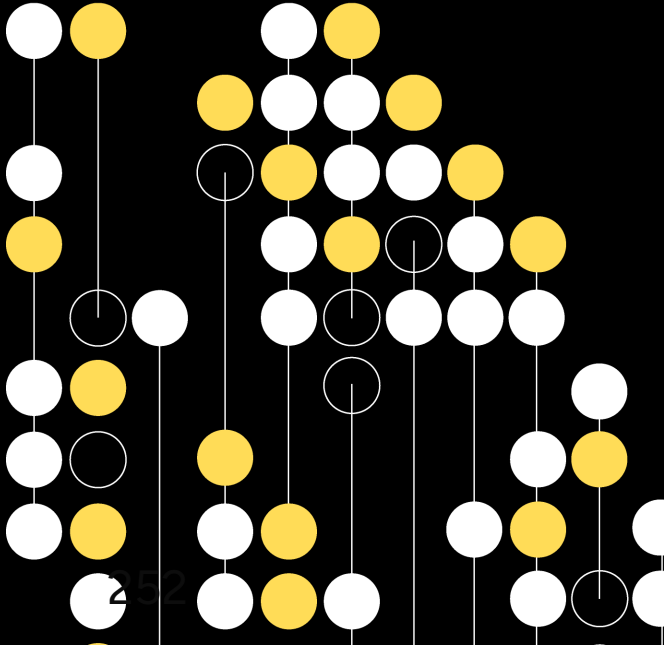
If you need help, feel free to reach out to your tutor.



Computer names and services: Summary

- **Computer names and services** are essential for creating user-friendly, efficient, and scalable networks.
- They simplify complex networking by providing human-readable identifiers and structured access to resources.
- This enables seamless communication and collaboration between users and devices, while ensuring reliability and scalability.
- Network services enable tasks like browsing and file sharing, using protocols such as HTTP, SMTP, and FTP.
- These services follow a client-server model and are accessed via unique port numbers (e.g., HTTP on port 80).

DHCP



DHCP

Lesson overview:

- **Recap from Module 3**
 - DHCP Introduction
 - DHCP Practical
- **Topics:**
 - DORA
 - DHCP Pool
 - Leased Time
 - Static Addressing

DHCP: DORA

The **DORA** process is how the Dynamic Host Configuration Protocol (DHCP) assigns an IP address to a device on a network.

DORA has four key steps:

1. Discover
2. Offer
3. Request
4. Acknowledge

DHCP: Discover in DORA (DHCPDISCOVER)

What is the Discover step?

- When a PC connects to the network the source IP address is 0.0.0.0, indicating the PC does not have an IP address yet.
- **The PC does not know** the IP address or MAC address of the DHCP server.
- The PC sends out a **broadcast message** with its MAC address and destination IP address set at 255.255.255.255 to all the devices present in that network.
- The PC communicates with the DHCP server using UDP port 68 as the source port and UDP port 67 as the destination port, which is why DHCP operates on ports 67 and 68.

Simple terms:

The device sends out a **broadcast** message asking for an IP address. This is the **Discover** step, where the device is saying, *"Who can give me an IP address?"*

DHCP: Offer in DORA (DHCPOFFER):

What is the Offer step?

- The next message is the DHCP Offer, sent by the DHCP server to the PC.
- It provides an IP address for the PC to use, along with other details such as the default gateway, DNS server, and more.
- The Offer is sent as a unicast frame to the PC's MAC address, since the server learned the PC's MAC address from the Discover message.

Simple terms:

It's like the DHCP server saying, *"Here's an IP address you can use."*

DHCP: Request in DORA (DHCPREQUEST)

What is the Request step?

- The PC responds to the DHCP server with a Request message.
- It confirms that it wants to use the offered IP address.
- This is important, there may be multiple DHCP servers on the local network.
- The source IP is still 0.0.0.0, since the offered IP address hasn't been configured yet.
- The PC sends out a **request** broadcast to accept the IP address offer from the server.

Simple terms:

This is the device saying, *"I'll take that IP address. Please assign it to me."*

DHCP: Acknowledge in DORA (DHCPACK):

What is the Acknowledge step?

- The DHCP server sends an Acknowledge message to confirm the assignment.
- The server sends an acknowledgement broadcast, along with the **lease time** to the DHCP client.
- It finalises the process and lets the device start using the IP address.

Simple terms:

It's the server saying, *"IP address assigned. You're good to go!"*

DHCP pool

- A collection of IP addresses managed by a DHCP server.
- When a device, such as a computer or smartphone, connects to the network, the server assigns an available IP address from this pool to the device.
- This ensures each device receives a unique address for communication without requiring manual configuration.
- The DHCP pool also provides additional settings to help devices automatically configure for the network. The following features ensure devices connect and operate seamlessly within the network:
 - Subnet Mask
 - Default Gateway
 - DNS Server
 - Lease Time
 - DHCP Options
 - Exclusion Ranges
 - IP Address Reservations
 - Network Time Protocol (NTP) Servers

DHCP options

- **Additional configuration settings sent to devices:**
 - **Option 43:** Custom vendor-specific information.
 - **Option 66/67:** Used for PXE boot, providing TFTP server info for network booting.
 - **Option 150:** Specifies VoIP configuration for IP phones.
- **Exclusion ranges:**
 - Defines specific IP addresses that the DHCP server will not assign.
 - Useful for reserving addresses for specific devices or services.
- **IP address reservations:**
 - Allows assigning a fixed IP address to a specific device based on its MAC address.
 - Ensures critical devices (e.g., printers, servers) always get the same IP.
- **Network Time Protocol (NTP) servers:**
 - Provides the address of NTP servers for clock synchronisation.

DHCP: Leased Time

- **DHCP Lease Time** refers to the duration a device is permitted to use a specific IP address assigned by a DHCP server.
- This lease period ensures efficient management of IP addresses within a network.

How DHCP Lease Time works:

- **Initial assignment:** When a device connects to the network, the DHCP server assigns an IP address along with a **lease time**.
- **Lease renewal:** Before the lease expires, the device attempts to renew the lease to continue using the same IP address.
- **Lease expiration:** If the device doesn't renew the lease in time, the IP address is returned to the pool for reassignment. The device could auto assign itself with an APIPA address at this point or revert to 0.0.0.0 to start the DORA process again.

DHCP: Leased Time T1

How DHCP Lease Time works:

- **At the DHCPACK stage:** The PC receives its leased IP address from the DHCP server, along with two timers (T1 and T2) that manage lease renewal and rebinding of IP addresses.

T1 Timer (renewal time):

- **Purpose:** Indicates when the client should first attempt to renew its lease with the original DHCP server.
- **Duration:** Typically set to 50% of the lease duration (e.g., for an 8-day lease, T1 is set to 4 days).
- **Process:**
 - The client sends a unicast **DHCPREQUEST** to the DHCP server.
 - If the server responds with a **DHCPACK**, the lease is renewed and the T1 timer is reset.
 - If no response is received, the client waits until T2 to try another renewal method.

DHCP: Leased Time T2

T2 Timer (rebinding time):

- **Purpose:** If the original DHCP server does not respond, T2 starts a broader attempt to rebind with any available DHCP server.

T2 Timer details:

- **Duration:** Typically set to 87.5% of the lease duration (e.g., for an 8-day lease, T2 occurs at 7 days).
- **Process:**
 - The client sends a broadcast **DHCPREQUEST**, seeking a response from any available DHCP server.
 - If a server responds, the lease is renewed, and both T1 and T2 timers are reset.
 - If no response is received, the lease expires, and the client will stop using the IP address, reverting to an **APIPA** or **0.0.0.0** address.

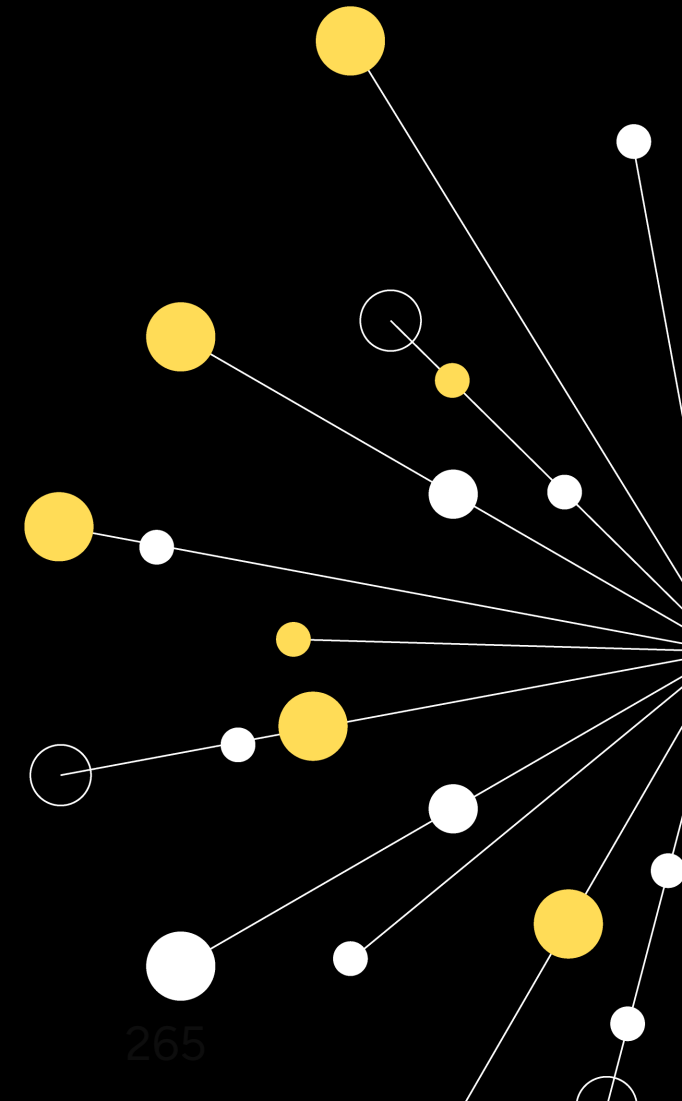
DHCP: Static IP addressing

- In static IP addressing, a device is manually configured with a fixed IP address that doesn't change.
- Manually assigned: The IP address is set on the device or router and does not change.
- Permanent address: Ideal for devices that require consistent connectivity.
- Reliable and consistent for devices like servers, printers, or cameras.
- Easier to configure for services requiring predictable IPs (e.g., hosting websites or file sharing).
- No reliance on a DHCP server.

Activity 28: DHCP

Complete the activity in your Exercise Guide. Once finished, use Miro to share your answers and findings.

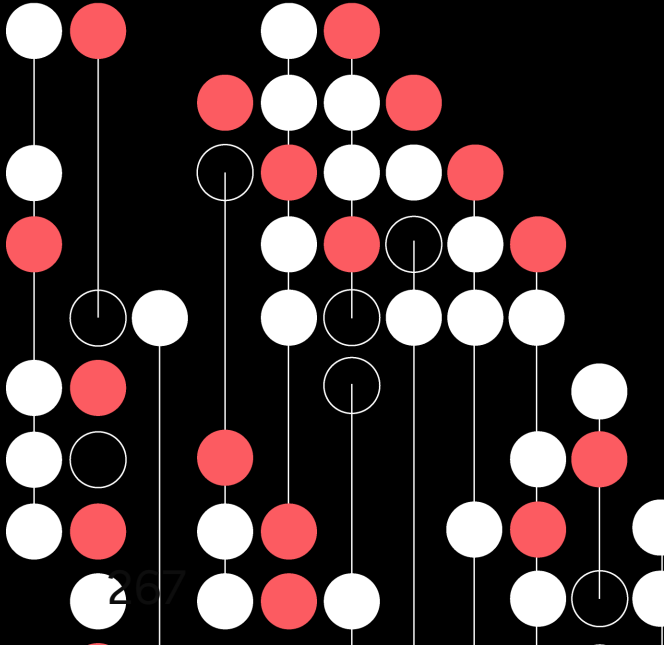
If you need help, feel free to reach out to your tutor.



DHCP: Summary

- DHCP automates the assignment of IP addresses and other configuration settings to devices on a network.
- By dynamically allocating resources, DHCP eliminates the need for manual configuration.
- DHCP simplifies network management, and ensures the efficient use of available IP addresses.
- Lease management ensures devices maintain proper IP addresses for a specified duration.
- Renewal (T1) and rebinding (T2) enable seamless connectivity in dynamic network environments.
- DHCP is foundational for both small-scale and enterprise networks, ensuring reliability, scalability, and ease of use, while supporting a wide range of devices and applications.

DNS



DNS

Lesson overview:

- **Recap from Module 3**
 - DNS Introduction
 - Public and Private DNS
 - DNS Practical
 - DNS Commands
- **Topics:**
 - FQDN
 - DNS Query
 - DNS Records

DNS: Fully Qualified Domain Name (FQDN)

Here is a complete domain name used to locate a specific website or service on the internet: www.example.com

The **TLD** (Top Level Domain) and the last part of the domain name (.com, .org, .net) indicate the type or category of the domain:

- **Generic TLD (gTLD):** Common TLDs like .com, .org, .net
- **Country-Code TLD (ccTLD):** Specific to countries, like .uk, .de, .ca
- **Sponsored TLD:** Restricted domains for specific industries, like .gov, .edu, .mil.

DNS: Fully Qualified Domain Name (FQDN)

Second-Level Domain (SLD):

- The part directly to the left of the TLD (**example** in example.com).
- It represents the name of the organisation, brand, or purpose of the website.
- Combined with the TLD, it creates the **primary domain** (example.com).

Subdomain:

- The part to the left of the SLD (**www** in www.example.com).
- Used to organise and separate different sections or services of a website.
- **blog**.example.com (for a blog) or **store**.example.com (for an e-commerce site).

DNS: Fully Qualified Domain Name (FQDN)

Root Domain:

- The combination of the SLD and TLD (**example.com**).
- This is the main address of the website without any subdomains.

Protocol:

- Not part of the domain but relevant: Preceding part like **http://** or **https://**.
- Indicates the method of communication between the browser and the server.

Breakdown of **blog.shop.example.co.uk**:

- TLD: **.uk** (Country-Code TLD for the United Kingdom).
- Second-Level Domain (SLD): **.co** (indicates commercial use).
- Root Domain: **example.co.uk** (primary domain name).
- Subdomain: **blog.shop** (subdomains organising different sections).
- FQDN: **blog.shop.example.co.uk** (complete domain name).

DNS: DNS query process

- User enters a **domain name** (www.example.com) in the browser.
- Browser sends a **DNS query** to resolve the domain into an IP address.
- OS/browser checks the **local DNS cache** for the IP address.
- Query goes to a **DNS Recursive Resolver** (ISP or third-party DNS like Google DNS).
- **Resolver** queries the **Root DNS server** for the TLD (.com, .org, .net).
- **TLD server** provides the location of the authoritative DNS server for the domain.
- **Resolver** queries the authoritative DNS server, which holds the IP address.
- **Authoritative server** responds with the IP address (192.0.2.1).
- **Recursive resolver** returns the IP address to the user's device.
- Browser uses the IP address to connect to the web server and load the website.

DNS: DNS Records

- Key elements of the Domain Name System (DNS).
- They store information about a domain name and direct how internet traffic should be handled for that domain.
- Each record serves a specific purpose, enabling services like website hosting, email delivery, and IP resolution.

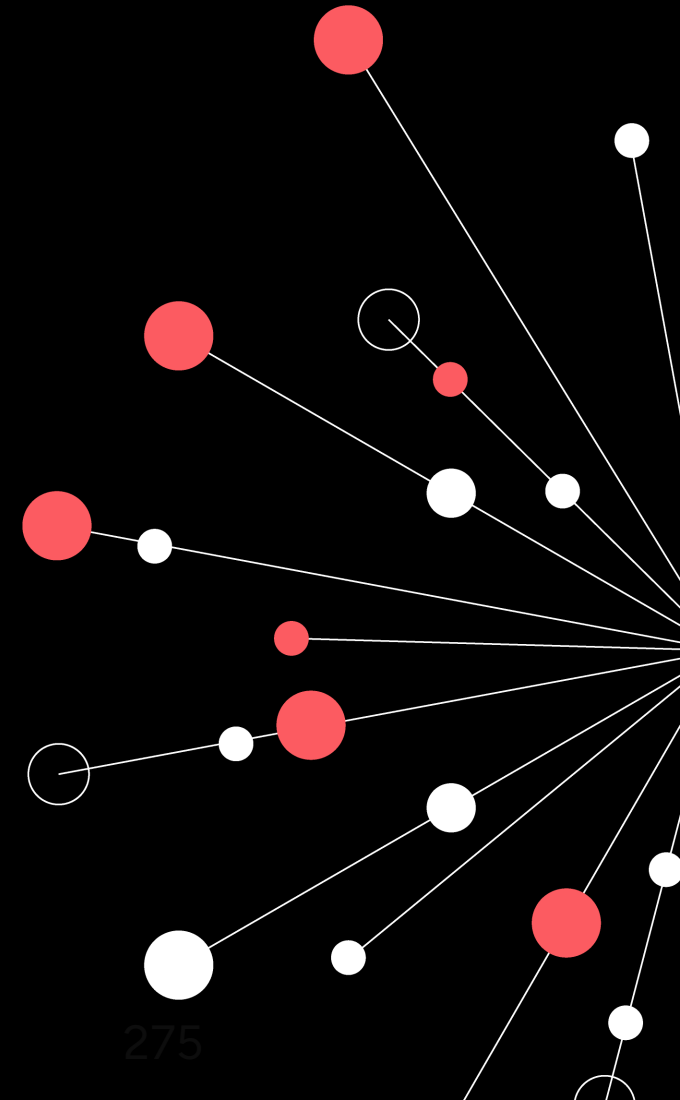
DNS: DNS Records

- **A Record** (Address Record) Maps a domain name to an IPv4 address: example.com → 192.0.2.1
- **AAAA Record** (IPv6 Address Record) Maps a domain name to an IPv6 address: example.com → 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- **CNAME Record** (Canonical Name Record) Points a domain or subdomain to another domain name (aliasing): www.hotmail.com → outlook.live.com
- **MX Record** (Mail Exchange Record) Directs email to the correct mail server for a domain: example.com → mail.example.com
- **NS Record** (Name Server Record) Specifies the authoritative name servers for a domain: example.com → ns1.example.com, ns2.example.com
- **PTR Record** (Pointer Record) Maps an IP address to a domain name (reverse DNS): 192.0.2.1 → example.com
- **SOA Record** (Start of Authority Record) Contains administrative information about a domain:
Primary name server.
Email of the domain administrator.
Serial number for DNS zone changes.
- **TXT Record** (Text Record) Stores human-readable text or machine-readable data for a domain. Often used for verifying domain ownership.

Activity 28: DNS

Complete the activity in your Exercise Guide. Once finished, use Miro to share your answers and findings.

If you need help, feel free to reach out to your tutor.



DNS: Summary

- Fully Qualified Domain Names (FQDNs), DNS queries, and DNS records form the backbone of modern internet communication.
- The FQDN provides a complete and unique name for every resource on the network, ensuring accurate identification and navigation.
- DNS queries act as the mechanism to resolve these names into machine-readable IP addresses, bridging the gap between human-friendly domain names and technical network addressing.
- DNS records, such as A, MX, CNAME, and TXT, store vital information about domains and their associated services, enabling efficient communication, email routing, and domain validation.

End of module review

- Computer Components
- Operating Systems
- Unix/Linux
- Microsoft Windows
- Apple
- Android
- Operating System Install
- Principles of Virtualisation
- Virtual Desktop Infrastructure
- Operating System Security and Updates
- System Tools and Resource Management
- Event Viewer and Log Files

End of module review

- Physical Components Of A Network
- Types of Networks
- Networking Models
- Wireless Networking
- Principles of Network Addressing
- IPV4
- IPv4 Address Allocation
- IPV6
- Computer Names and Services
- Network Services
- DHCP
- DNS

End of module review

Kindly upload your completed Delegate Guide and
*Certification of Completion to BUD under the following
module:

- Activity (Live Event)
- *Certification of Completion can be obtained by going to the live event module in BUD and pressing "**start now**", then tick off the modules accordingly.

Learning instructions & materials

To complete this activity:

1. Attend a two-day live event (*), led by your tutor | [Start now](#) | - please don't start this learning path before the live event.
2. Submit your work in the 'Messages and Submissions' area above

